复旦数学资料汇编
量子场论

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第一章　量子场论

1.1　习题: 量子场论

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comment| 这个习题解答是 2025 秋 SIMIS Jean-Emile Bourgine 老师量子场论（一）课程的习题，每次占比 10%.

1.2　Assignment 1

Exercise 1. *(10 points) In this exercise, we want to study the action for gravitational waves. Our starting point is the Einstein equations for the gravitational field $g_{\mu\nu}(x)$ identified with the spacetime metric,*[1]

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 0, \tag{1.2}$$

where we used the following notations:

- **Ricci scalar:** $R = g^{\mu\nu} R_{\mu\nu}$,
- **Ricci tensor:** $R_{\mu\nu} = R^{\rho}_{\mu\rho\nu}$,
- **Riemann tensor:** $R^{\rho}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\sigma\nu} - \partial_\nu \Gamma^{\rho}_{\sigma\mu} + \Gamma^{\rho}_{\mu\alpha}\Gamma^{\alpha}_{\nu\sigma} - \Gamma^{\rho}_{\nu\alpha}\Gamma^{\alpha}_{\mu\sigma}$,
- **Christoffel symbols:** $\Gamma^{\mu}_{\nu\rho} = \frac{1}{2} g^{\mu\sigma}(\partial_\rho g_{\sigma\nu} + \partial_\nu g_{\sigma\rho} - \partial_\sigma g_{\nu\rho})$.

The metric $g_{\mu\nu}(x)$ is assumed symmetric and non-degenerate, and we denote $g^{\mu\nu}(x)$ the inverse metric, i.e.

$$g^{\mu\nu}(x) g_{\nu\rho}(x) = \delta^{\mu}_{\rho} = g_{\rho\nu}(x) g^{\nu\mu}(x). \tag{1.3}$$

1. We consider a small perturbation around the flat metric $g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x)$, derive the linearized equations of motion (at order in $O(h)$). *(2 points)*
2. Let $h = h^{\mu}_{\mu} = \eta^{\mu\nu} h_{\mu\nu}$, we consider the Fierz-Pauli action for the symmetric tensor $h_{\mu\nu}(x)$,

$$S[h] = \int d^4x \left[-\frac{1}{4} \partial_\rho h_{\mu\nu} \partial^\rho h^{\mu\nu} + \frac{1}{2} \partial_\rho h_{\mu\nu} \partial^\nu h^{\rho\mu} + \frac{1}{4} \partial^\rho h \partial_\rho h - \frac{1}{2} \partial_\rho h \partial_\mu h^{\mu\rho} \right] \tag{1.4}$$

[1] We use here the standard notation for general relativity, the flat metric will be denoted

$$\eta^{\mu\nu} = \eta_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \tag{1.1}$$

while the notation $g_{\mu\nu}(x)$ is reserved for the dynamical field describing the metric.

Show that the Euler-Lagrange equations reproduce the previous linearized equations of motion for $h_{\mu\nu}$ *(2 points)*.

3. Compute the fields $\Pi_{\mu\nu}$ canonically conjugate to $h_{\mu\nu}$. *(1 point)*
4. Compute the energy-momentum tensor associated to the Fierz-Pauli action. Deduce the Hamiltonian. *(2 points)*
5. Show that the theory is invariant under the transformation $h_{\mu\nu}(x) \to h_{\mu\nu}(x) + \partial_\mu V_\nu(x) + \partial_\nu V_\mu(x)$. *(3 points)*

Solution:

1. We compute the expansion at order $O(h)$ of the Ricci scalar and Ricci tensor,

$$\begin{aligned}
\Gamma^\mu_{\nu\rho} &= \frac{1}{2}\eta^{\mu\sigma}(\partial_\rho h_{\sigma\nu} + \partial_\nu h_{\sigma\rho} - \partial_\sigma h_{\nu\rho}) + O(h^2),\\
R_{\mu\nu\rho\sigma} &= \frac{1}{2}\left(\partial_\rho\partial_\nu h_{\mu\sigma} + \partial_\mu\partial_\sigma h_{\rho\nu} - \partial_\rho\partial_\mu h_{\nu\sigma} - \partial_\nu\partial_\sigma h_{\rho\mu}\right) + O(h^2),\\
R_{\mu\nu} &= \frac{1}{2}\left(\partial_\sigma\partial_\nu h^\sigma_\mu + \partial_\sigma\partial_\mu h^\sigma_\nu - \partial_\mu\partial_\nu h - \Box h_{\mu\nu}\right) + O(h^2),\\
R &= \partial_\mu\partial_\nu h^{\mu\nu} - \Box h,
\end{aligned} \tag{1.5}$$

with $h = h^\mu_\mu$. Thus, at first order, Einstein's equation becomes

$$\partial_\sigma\partial_\nu h^\sigma_\mu + \partial_\sigma\partial_\mu h^\sigma_\nu - \partial_\mu\partial_\nu h - \Box h_{\mu\nu} - \eta_{\mu\nu}\partial_\rho\partial_\sigma h^{\rho\sigma} + \eta_{\mu\nu}\Box h = 0. \tag{1.6}$$

2. Let's rewrite the Lagrangian density as follows,

$$\mathcal{L} = \eta^{\mu\mu'}\eta^{\nu\nu'}\eta^{\rho\rho'}\left(-\frac{1}{4}\partial_\rho h_{\mu\nu}\partial_{\rho'}h_{\mu'\nu'} + \frac{1}{2}\partial_\rho h_{\mu\nu}\partial_{\nu'}h_{\rho'\mu'} + \frac{1}{4}\partial_{\rho'}h_{\mu\mu'}\partial_\rho h_{\nu\nu'} - \frac{1}{2}\partial_\rho h_{\nu\nu'}\partial_\mu h_{\mu'\rho'}\right) \tag{1.7}$$

Obviously, we have $\partial\mathcal{L}/\partial h_{\alpha\beta} = 0$, so let's compute

$$\begin{aligned}
\frac{\partial\mathcal{L}}{\partial(\partial_\gamma h_{\alpha\beta})} &= -\frac{1}{4}(\partial^\gamma h^{\alpha\beta} + \partial^\gamma h^{\alpha\beta}) + \frac{1}{2}(\partial^\beta h^{\gamma\alpha} + \partial^\alpha h^{\beta\gamma}) + \frac{1}{4}(\eta^{\alpha\beta}\partial^\gamma h + \eta^{\alpha\beta}\partial^\gamma h) - \frac{1}{2}(\eta^{\alpha\beta}\partial_\mu h^{\mu\gamma} + \eta^{\gamma\alpha}\partial^\beta h)\\
&= -\frac{1}{2}\partial^\gamma h^{\alpha\beta} + \frac{1}{2}\partial^\beta h^{\gamma\alpha} + \frac{1}{2}\partial^\alpha h^{\beta\gamma} + \frac{1}{2}\eta^{\alpha\beta}\partial^\gamma h - \frac{1}{2}\eta^{\alpha\beta}\partial_\mu h^{\mu\gamma} - \frac{1}{2}\eta^{\alpha\gamma}\partial^\beta h.
\end{aligned} \tag{1.8}$$

The Euler-Lagrange equation

$$\partial_\gamma\frac{\partial\mathcal{L}}{\partial(\partial_\gamma h_{\alpha\beta})} = 0 \tag{1.9}$$

gives

$$-\frac{1}{2}\Box h^{\alpha\beta} + \frac{1}{2}\partial_\gamma\partial^\beta h^{\gamma\alpha} + \frac{1}{2}\partial_\gamma\partial^\alpha h^{\beta\gamma} + \frac{1}{2}\eta^{\alpha\beta}\Box h - \frac{1}{2}\eta^{\alpha\beta}\partial_\gamma\partial_\sigma h^{\sigma\gamma} - \frac{1}{2}\partial^\alpha\partial^\beta h = 0 \tag{1.10}$$

Let's apply $2\eta_{\alpha\mu}\eta_{\beta\nu}$ to this equation, using the fact that $h_{\mu\nu}$ is symmetric we recover

$$-\Box h_{\mu\nu} + \partial_\sigma\partial_\nu h^\sigma_\mu + \partial_\sigma\partial_\mu h^\sigma_\nu + \eta_{\mu\nu}\Box h - \eta_{\mu\nu}\partial_\rho\partial_\sigma h^{\rho\sigma} - \partial_\mu\partial_\nu h = 0. \tag{1.11}$$

3. It is just a matter of applying the formula,

$$\Pi^{\alpha\beta} = \frac{\partial\mathcal{L}}{\partial\dot{h}_{\alpha\beta}} = -\frac{1}{2}\dot{h}^{\alpha\beta} + \frac{1}{2}(\partial^\alpha h^{0\beta} + \partial^\beta h^{0\alpha}) + \frac{1}{2}\eta^{\alpha\beta}(\dot{h} - \partial_\mu h^{0\mu}) - \frac{1}{2}\eta^{\alpha0}\partial^\beta h. \tag{1.12}$$

4. Again, we just apply the formula,

$$\begin{aligned} T^{\mu\nu} &= \frac{\partial \mathcal{L}}{\partial(\partial_\mu h_{\alpha\beta})}\partial^\nu h_{\alpha\beta} - \mathcal{L}\eta^{\mu\nu} \\ &= \frac{1}{2}(-\partial^\mu h^{\alpha\beta}\partial^\nu h_{\alpha\beta} + 2\partial^\alpha h^{\mu\beta}\partial^\nu h_{\alpha\beta} + \partial^\mu h\partial^\nu h - \partial_\rho h^{\mu\rho}\partial^\nu h - \partial^\alpha h\partial^\nu h^\mu_\alpha) - \mathcal{L}\eta^{\mu\nu}. \end{aligned} \tag{1.13}$$

and

$$\mathcal{H} = T^{00} - \mathcal{L} = -\frac{1}{2}\dot{h}^{\alpha\beta}\dot{h}_{\alpha\beta} + \dot{h}_{\alpha\beta}\partial^\alpha h^{0\beta} + \frac{1}{2}\dot{h}^2 - \frac{1}{2}\dot{h}\partial_\alpha h^{0\alpha} - \dot{h}^{\mu\alpha}\partial_\alpha h - \mathcal{L}. \tag{1.14}$$

5. There are two options, either show that the equations of motions are invariant (easier), or show that the Lagrangian is invariant up to a total divergence. We will take the second option here, which is a rather tedious calculation. For simplicity, we split the Lagrangian density into two parts,

$$\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_2, \quad \mathcal{L}_1 = \frac{1}{2}\partial_\rho h_{\mu\nu}\left(\partial^\nu h^{\rho\mu} - \frac{1}{2}\partial^\rho h^{\mu\nu}\right), \quad \mathcal{L}_2 = \frac{1}{2}\partial_\rho h\left(\frac{1}{2}\partial^\rho h - \partial_\mu h^{\mu\rho}\right). \tag{1.15}$$

Let's start with the terms linear in V, after some cancelations, we find the following variations,

$$\begin{aligned} \delta\mathcal{L}_1^{(1)} &= \partial_\rho h^{\mu\nu}\partial_\mu\partial_\nu V^\rho = \partial_\rho(h^{\mu\nu}\partial_\mu\partial_\nu V^\rho) - h^{\mu\nu}\partial_\mu\partial_\nu\partial_\rho V^\rho, \\ \delta\mathcal{L}_2^{(1)} &= \frac{1}{2}\partial^\rho h\partial_\rho\partial_\mu V^\mu - \frac{1}{2}\partial^\rho h\Box V_\rho - \partial_\mu(h^{\mu\rho})\partial_\rho\partial_\nu V^\nu, \\ &= \frac{1}{2}\left(\partial^\rho(h\partial_\rho\partial_\mu V^\mu) - h\partial_\mu\Box V^\mu\right) - \frac{1}{2}\left(\partial^\rho(h\Box V_\rho) - h\Box\partial^\rho V_\rho\right) - \partial_\mu(h^{\mu\rho}\partial_\rho\partial_\nu V^\nu) + h^{\mu\rho}\partial_\mu\partial_\rho\partial_\nu V^\nu \\ &= \frac{1}{2}\partial_\rho\left[h\partial^\rho\partial_\mu V^\mu - h\Box V^\rho - 2h^{\mu\rho}\partial_\mu\partial_\nu V^\nu\right] + h^{\mu\nu}\partial_\mu\partial_\nu\partial_\rho V^\rho. \end{aligned} \tag{1.16}$$

Thus, $\delta\mathcal{L}_1^{(1)} + \delta\mathcal{L}_2^{(1)}$ is a divergence. Next, we examine the quadratic terms in V,

$$\begin{aligned} \delta\mathcal{L}_1^{(2)} &= \frac{1}{2}\partial_\rho(\partial_\mu V_\nu + \partial_\nu V_\mu)\left(\partial^\nu\partial^\mu V^\rho + \partial^\nu\partial^\rho V^\mu - \partial^\rho\partial^\mu V^\nu\right) = \partial_\mu\partial_\nu V_\rho\partial^\rho\partial^\mu V^\nu, \\ \delta\mathcal{L}_2^{(2)} &= \partial_\rho\partial_\mu V^\mu(\partial^\rho\partial^\nu V_\nu - \partial_\nu\partial^\nu V^\rho - \partial_\nu\partial^\rho V^\nu) = -\partial_\rho\partial_\mu V^\mu\partial_\nu\partial^\nu V^\rho \\ &= -\partial_\nu\left[\partial^\nu V^\rho\partial_\rho\partial_\mu V^\mu\right] + \partial^\nu V^\rho\partial_\nu\partial_\rho\partial_\mu V^\mu, \\ &= -\partial_\nu\left[\partial^\nu V^\rho\partial_\rho\partial_\mu V^\mu\right] + \partial_\mu\left[\partial_\nu\partial_\rho V^\mu\partial^\nu V^\rho\right] - \partial_\nu\partial_\rho V^\mu\partial_\mu\partial^\nu V^\rho, \\ &= \partial_\rho\left[-\partial^\rho V^\mu\partial_\mu\partial_\nu V^\nu + \partial_\mu\partial_\nu V^\rho\partial^\mu V^\nu\right] - \partial_\mu\partial_\nu V_\rho\partial^\rho\partial^\mu V^\nu. \end{aligned} \tag{1.17}$$

We observe that $\delta\mathcal{L}_1^{(2)} + \delta\mathcal{L}_2^{(2)}$ is again a total divergence, which concludes the proof.

Exercise 2. *(10 points) We consider a massive spin 1/2 particle described at $t = 0$ by a wave packet with a Gaussian distribution centered at $\mathbf{x} = 0$ and of half-width d,*

$$\Psi(0,\mathbf{x}) = \frac{1}{(\pi d^2)^{3/4}}e^{-\mathbf{x}^2/(2d^2)}\begin{pmatrix}\frac{\sqrt{2}}{2}\\ \frac{\sqrt{2}}{2}\\ 0\\ 0\end{pmatrix}. \tag{1.18}$$

1. Find the corresponding solution of the Dirac equation $\Psi(t,\mathbf{x})$. *(2 points)*
2. Compute the norm of the solution $\int d^3\mathbf{x}\Psi^\dagger(t,\mathbf{x})\Psi(t,\mathbf{x})$, and verify that it is time independent. *(3 points)*
3. Is there a limit in which we can neglect the negative energy solutions? *(1 points)*

4. Compute the following integrated current for the Gaussian wave packet,

$$j^\mu(t) = \int d^3\mathbf{x}\bar{\Psi}(t,\mathbf{x})\gamma^\mu\Psi(t,\mathbf{x}) \tag{1.19}$$

and show that its space components vanish, $j^i(t) = 0$. *(4 points)*

Solution: 1. A general solution of the Dirac equation has the form

$$\Psi(t,\mathbf{x}) = \int \frac{d^3\mathbf{p}}{(2\pi)^3}\sum_{r=1,2}\left[b_r(\mathbf{p})u_r(\mathbf{p})e^{-ipx} + d_r(\mathbf{p})v_r(\mathbf{p})e^{ipx}\right], \tag{1.20}$$

with the coefficients $b_r(\mathbf{p}), d_r(\mathbf{p})$ to be determined, and $u_r(\mathbf{p}), v_r(\mathbf{p})$ given by the formulas (2.44) of the lecture notes, namely

$$u_r(\mathbf{p}) = \frac{1}{\sqrt{p^0+m}}\begin{pmatrix}(p^0+m)\psi_r\\ \mathbf{p}\cdot\boldsymbol{\sigma}\psi_r\end{pmatrix}, \quad v_r(\mathbf{p}) = \frac{1}{\sqrt{p^0+m}}\begin{pmatrix}\mathbf{p}\cdot\boldsymbol{\sigma}\psi_r\\ (p^0+m)\psi_r\end{pmatrix}, \quad \text{with} \quad \psi_1 = \begin{pmatrix}1\\0\end{pmatrix}, \quad \psi_2 = \begin{pmatrix}0\\1\end{pmatrix}. \tag{1.21}$$

The solution is on mass shell, and so $p^0 = p^0(\mathbf{p}) = \sqrt{\mathbf{p}^2+m^2}$ in this formula. Taking the Fourier transform of the wave function at initial time, we find

$$\int d^3\mathbf{x}e^{-i\mathbf{p}\cdot\mathbf{x}}\Psi(0,\mathbf{x}) = (4\pi d^2)^{3/4}e^{-\mathbf{p}^2d^2/2}\Xi, \quad \Xi = \frac{\sqrt{2}}{2}\begin{pmatrix}1\\1\\0\\0\end{pmatrix}. \tag{1.22}$$

Comparing with the Fourier transform of the general solution at $t = 0$, we find the condition

$$(4\pi d^2)^{3/4}e^{-\mathbf{p}^2d^2/2}\Xi = \sum_{r=1,2}[b_r(\mathbf{p})u_r(\mathbf{p}) + d_r(-\mathbf{p})v_r(-\mathbf{p})]. \tag{1.23}$$

To solve this equation, we use the projection relations (2.45) of the lecture notes, it gives

$$b_r(\mathbf{p}) = \frac{(4\pi d^2)^{3/4}}{2p^0}e^{-\mathbf{p}^2d^2/2}u_r(\mathbf{p})^\dagger\Xi, \quad d_r(-\mathbf{p}) = \frac{(4\pi d^2)^{3/4}}{2p^0}e^{-\mathbf{p}^2d^2/2}v_r(-\mathbf{p})^\dagger\Xi. \tag{1.24}$$

Introducing

$$\xi = \frac{\sqrt{2}}{2}\begin{pmatrix}1\\1\end{pmatrix} \quad \Rightarrow \quad \Xi = \begin{pmatrix}\xi\\0\end{pmatrix}, \tag{1.25}$$

we deduce (recall that $\boldsymbol{\sigma}^\dagger = \boldsymbol{\sigma}$),[2]

$$\begin{aligned} b_r(\mathbf{p}) &= \frac{(4\pi d^2)^{3/4}}{2p^0}e^{-\mathbf{p}^2d^2/2}\frac{1}{\sqrt{p^0+m}}(p^0+m)\psi_r^\dagger\xi = \frac{(4\pi d^2)^{3/4}}{2p^0}e^{-\mathbf{p}^2d^2/2}\sqrt{p^0+m}\frac{\sqrt{2}}{2},\\ d_r(\mathbf{p}) &= \frac{(4\pi d^2)^{3/4}}{2p^0}e^{-\mathbf{p}^2d^2/2}\frac{1}{\sqrt{p^0+m}}\psi_r^\dagger(\mathbf{p}\cdot\boldsymbol{\sigma})\xi,\\ \Rightarrow \quad d_1(\mathbf{p}) &= \frac{(4\pi d^2)^{3/4}}{2p^0}e^{-\mathbf{p}^2d^2/2}\frac{\sqrt{2}}{2}\frac{p^1-ip^2+p^3}{\sqrt{p^0+m}}, \quad d_2(\mathbf{p}) = \frac{(4\pi d^2)^{3/4}}{2p^0}e^{-\mathbf{p}^2d^2/2}\frac{\sqrt{2}}{2}\frac{p^1+ip^2-p^3}{\sqrt{p^0+m}}. \end{aligned} \tag{1.27}$$

[2]We used

$$\mathbf{p}\cdot\boldsymbol{\sigma} = \begin{pmatrix}p^3 & p^1-ip^2\\ p^1+ip^2 & -p^3\end{pmatrix} \quad \Rightarrow \quad \mathbf{p}\cdot\boldsymbol{\sigma}\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}p^1-ip^2+p^3\\ p^1+ip^2-p^3\end{pmatrix} \tag{1.26}$$

2. Let's denote

$$\mathcal{N} = \int d^3\mathbf{x}\Psi^\dagger(t,\mathbf{x})\Psi(t,\mathbf{x}) \tag{1.28}$$

We have

$$\begin{aligned}\mathcal{N} = \int d^3\mathbf{x}\int\frac{d^3\mathbf{p}_1}{(2\pi)^3}\int\frac{d^3\mathbf{p}_2}{(2\pi)^3}\sum_{r,s=1}^{2}\Big[&e^{ix(p_1-p_2)}b_r^*(\mathbf{p}_1)b_s(\mathbf{p}_2)u_r(\mathbf{p}_1)^\dagger u_s(\mathbf{p}_2)\\&+e^{-ix(p_1-p_2)}d_r^*(\mathbf{p}_1)d_s(\mathbf{p}_2)v_r(\mathbf{p}_1)^\dagger v_s(\mathbf{p}_2)\\&+e^{ix(p_1+p_2)}b_r^*(\mathbf{p}_1)d_s(\mathbf{p}_2)u_r(\mathbf{p}_1)^\dagger v_s(\mathbf{p}_2)\\&+e^{-ix(p_1+p_2)}d_r^*(\mathbf{p}_1)b_s(\mathbf{p}_2)v_r(\mathbf{p}_1)^\dagger u_s(\mathbf{p}_2)\Big]\end{aligned} \tag{1.29}$$

The integral over $d^3\mathbf{x}$ produces delta functions, e.g.

$$\begin{aligned}\int d^3\mathbf{x}e^{-ix(p_1+p_2)} &= e^{-it(p^0(\mathbf{p}_1)+p^0(\mathbf{p}_2))}\int d^3\mathbf{x}e^{i\mathbf{x}\cdot(\mathbf{p}_1+\mathbf{p}_2)} = (2\pi)^3\delta(\mathbf{p}_1+\mathbf{p}_2)e^{-2itp^0(\mathbf{p}_1)}\\ \int d^3\mathbf{x}e^{-ix(p_1-p_2)} &= e^{-it(p^0(\mathbf{p}_1)-p^0(\mathbf{p}_2))}\int d^3\mathbf{x}e^{i\mathbf{x}\cdot(\mathbf{p}_1-\mathbf{p}_2)} = (2\pi)^3\delta(\mathbf{p}_1-\mathbf{p}_2).\end{aligned} \tag{1.30}$$

since $p^0(\mathbf{p}) = p^0(-\mathbf{p})$. This gives

$$\begin{aligned}\mathcal{N} = \int\frac{d^3\mathbf{p}}{(2\pi)^3}\sum_{r,s=1}^{2}\Big[&b_r^*(\mathbf{p})b_s(\mathbf{p})u_r(\mathbf{p})^\dagger u_s(\mathbf{p}) + d_r^*(\mathbf{p})d_s(\mathbf{p})v_r(\mathbf{p})^\dagger v_s(\mathbf{p})\\&+e^{2itp^0}b_r^*(\mathbf{p})d_s(-\mathbf{p})u_r(\mathbf{p})^\dagger v_s(-\mathbf{p}) + e^{-2itp^0}d_r^*(\mathbf{p})b_s(-\mathbf{p})v_r(\mathbf{p})^\dagger u_s(-\mathbf{p})\Big]\end{aligned} \tag{1.31}$$

Next, we use the projection relations (2.45), namely[3]

$$u_r^\dagger(\mathbf{p})u_s(\mathbf{p}) = v_r^\dagger(\mathbf{p})v_s(\mathbf{p}) = 2p^0\delta_{r,s},\quad u_r^\dagger(\mathbf{p})v_s(-\mathbf{p}) = v_r^\dagger(\mathbf{p})u_s(-\mathbf{p}) = 0, \tag{1.33}$$

Most terms disappear, there only remains

$$\mathcal{N} = \int\frac{d^3\mathbf{p}}{(2\pi)^3}2p^0\sum_{r=1}^{2}\left[b_r^*(\mathbf{p})b_r(\mathbf{p}) + d_r^*(\mathbf{p})d_r(\mathbf{p})\right]. \tag{1.34}$$

This is indeed time independent. To compute the remaining sum, we recall the previous formulas for the coefficients

$$b_r(\mathbf{p}) = F(\mathbf{p})(p^0+m)\xi_r,\quad d_r(\mathbf{p}) = F(\mathbf{p})(\mathbf{p}\cdot\boldsymbol{\sigma}\xi)_r,\quad \text{with}\quad F(\mathbf{p}) = \frac{(4\pi d^2)^{3/4}}{2p^0}\frac{e^{-\mathbf{p}^2d^2/2}}{\sqrt{p^0+m}}. \tag{1.35}$$

Note that $F(\mathbf{p}) = F(-\mathbf{p}) = F^*(\mathbf{p})$. Thus, we have

$$\begin{aligned}\sum_{r=1}^{2}\left[|b_r(\mathbf{p})|^2+|d_r(\mathbf{p})|^2\right] &= F(\mathbf{p})^2\left[(p^0+m)^2\xi^\dagger\xi + \xi^\dagger(\mathbf{p}\cdot\boldsymbol{\sigma})^2\xi\right] = F(\mathbf{p})^2\left[(p^0+m)^2+\mathbf{p}^2\right]\\&= 2p^0(p^0+m)F(\mathbf{p})^2 = \frac{(4\pi d^2)^{3/2}}{2p^0}e^{-\mathbf{p}^2d^2},\end{aligned} \tag{1.36}$$

[3]This can also be checked directly using $\psi_r^\dagger\psi_s = \delta_{r,s}$ and $(\mathbf{p}\cdot\boldsymbol{\sigma})^2 = \frac{1}{2}p^ip^j\{\sigma^i,\sigma^j\} = \mathbf{p}^2$, e.g.

$$\begin{aligned}u_r^\dagger(\mathbf{p})u_s(\mathbf{p}) &= \frac{1}{p^0+m}\left[(p^0+m)^2\psi_r^\dagger\psi_s + \psi_r^\dagger(\mathbf{p}\cdot\boldsymbol{\sigma})^2\psi_s\right] = \frac{\delta_{r,s}}{p^0+m}\left[(p^0+m)^2+\mathbf{p}^2\right] = 2p^0\delta_{r,s},\\ u_r^\dagger(\mathbf{p})v_s(-\mathbf{p}) &= \frac{1}{p^0+m}\left[-(p^0+m)\psi_r^\dagger(\mathbf{p}\cdot\boldsymbol{\sigma})\psi_s + (p^0+m)\psi_r^\dagger(\mathbf{p}\cdot\boldsymbol{\sigma})\psi_s\right] = 0.\end{aligned} \tag{1.32}$$

using $(\mathbf{p}\cdot\boldsymbol{\sigma})^2=\mathbf{p}^2$ and $\xi^\dagger\xi=1$. Inserting the coefficients obtained before, we find

$$\mathcal{N}=(4\pi d^2)^{3/2}\int\frac{d^3\mathbf{p}}{(2\pi)^3}e^{-\mathbf{p}^2d^2}=(4\pi d^2)^{3/2}\left(\int_{-\infty}^{\infty}\frac{dp}{2\pi}e^{-p^2d^2}\right)^3=1. \tag{1.37}$$

3. In the non-relativistic limit $|\mathbf{p}|\ll m$, the coefficient in front of the negative energy solution becomes negligible since

$$\frac{d_r(\mathbf{p})}{b_s(\mathbf{p})}=\frac{(\mathbf{p}\cdot\boldsymbol{\sigma}\xi)_r}{p^0+m}\simeq\frac{p^i}{m}\ll 1. \tag{1.38}$$

4. First, we notice that

$$j^\mu=\int d^3\mathbf{x}\bar\Psi(t,\mathbf{x})\gamma^\mu\Psi(t,\mathbf{x})=\int d^3\mathbf{x}\Psi^\dagger(t,\mathbf{x})\gamma^0\gamma^\mu\Psi(t,\mathbf{x}) \tag{1.39}$$

and so $j^0=\mathcal{N}$, which indeed reproduces the formula. Thus, we only need to check the case $\mu=i$. The calculation starts in the same way as before, and using the delta functions, we end up with

$$\begin{aligned}j^\mu=\int\frac{d^3\mathbf{p}}{(2\pi)^3}\sum_{r,s=1}^{2}\Big[&b_r^*(\mathbf{p})b_s(\mathbf{p})\bar u_r(\mathbf{p})\gamma^\mu u_s(\mathbf{p})+d_r^*(\mathbf{p})d_s(\mathbf{p})\bar v_r(\mathbf{p})\gamma^\mu v_s(\mathbf{p})\\&+e^{2itp^0}b_r^*(\mathbf{p})d_s(-\mathbf{p})\bar u_r(\mathbf{p})\gamma^\mu v_s(-\mathbf{p})+e^{-2itp^0}d_r^*(\mathbf{p})b_s(-\mathbf{p})\bar v_r(\mathbf{p})\gamma^\mu u_s(-\mathbf{p})\Big]\end{aligned} \tag{1.40}$$

Let's decompose $j^\mu=j^\mu_{\text{reg}}+j^\mu_{\text{osc}}$ with

$$\begin{aligned}j^\mu_{\text{reg}}&=\int\frac{d^3\mathbf{p}}{(2\pi)^3}\sum_{r,s=1}^{2}\left[b_r^*(\mathbf{p})b_s(\mathbf{p})\bar u_r(\mathbf{p})\gamma^\mu u_s(\mathbf{p})+d_r^*(\mathbf{p})d_s(\mathbf{p})\bar v_r(\mathbf{p})\gamma^\mu v_s(\mathbf{p})\right]\\j^\mu_{\text{osc}}&=\int\frac{d^3\mathbf{p}}{(2\pi)^3}\sum_{r,s=1}^{2}\left[e^{2itp^0}b_r^*(\mathbf{p})d_s(-\mathbf{p})\bar u_r(\mathbf{p})\gamma^\mu v_s(-\mathbf{p})+e^{-2itp^0}d_r^*(\mathbf{p})b_s(-\mathbf{p})\bar v_r(\mathbf{p})\gamma^\mu u_s(-\mathbf{p})\right],\end{aligned} \tag{1.41}$$

since we have seen that $j^0_{\text{osc}}=0$. We need to compute these expressions. In Dirac's representation, we have the following expression

$$\gamma^0\gamma^i=\begin{pmatrix}0&\sigma^i\\\sigma^i&0\end{pmatrix}. \tag{1.42}$$

Using this expression, we derive[4]

$$\begin{aligned}\bar u_r(\mathbf{p})\gamma^i u_s(\mathbf{p})&=\bar v_r(\mathbf{p})\gamma^i v_s(\mathbf{p})=2p^i\delta_{r,s},\\\bar u_r(\mathbf{p})\gamma^i v_s(-\mathbf{p})&=\bar v_r(\mathbf{p})\gamma^i u_s(-\mathbf{p})=2p^0\sigma^i_{r,s}-\frac{2p^i}{p^0+m}\mathbf{p}\cdot\boldsymbol{\sigma}_{r,s}.\end{aligned} \tag{1.44}$$

[4]For instance, we have

$$\begin{aligned}\bar u_r(\mathbf{p})\gamma^i u_s(\mathbf{p})&=\frac{1}{p^0+m}\begin{pmatrix}(p^0+m)\psi_r^\dagger&\psi_r^\dagger(\mathbf{p}\cdot\boldsymbol{\sigma})\end{pmatrix}\cdot\begin{pmatrix}\sigma^i(\mathbf{p}\cdot\boldsymbol{\sigma})\psi_s\\(p^0+m)\sigma^i\psi_s\end{pmatrix}=p^j\psi_r^\dagger\left(\sigma^i\sigma^j+\sigma^j\sigma^i\right)\psi_s=2p^i\delta_{r,s},\\\bar u_r(\mathbf{p})\gamma^i v_s(-\mathbf{p})&=\frac{1}{p^0+m}\begin{pmatrix}(p^0+m)\psi_r^\dagger&\psi_r^\dagger(\mathbf{p}\cdot\boldsymbol{\sigma})\end{pmatrix}\cdot\begin{pmatrix}(p^0+m)\sigma^i\psi_s\\-\sigma^i(\mathbf{p}\cdot\boldsymbol{\sigma})\psi_s\end{pmatrix}\\&=\frac{1}{p^0+m}\left[(p^0+m)^2\psi_r^\dagger\sigma^i\psi_s-\psi_r^\dagger(\mathbf{p}\cdot\boldsymbol{\sigma})\sigma^i(\mathbf{p}\cdot\boldsymbol{\sigma})\psi_s\right]=\frac{1}{p^0+m}\left[((p^0+m)^2+\mathbf{p}^2)\psi_r^\dagger\sigma^i\psi_s-2p^i\psi_r^\dagger(\mathbf{p}\cdot\boldsymbol{\sigma})\psi_s\right]\\&=2p^0\sigma^i_{r,s}-\frac{2p^i}{p^0+m}\mathbf{p}\cdot\boldsymbol{\sigma}_{r,s}.\end{aligned} \tag{1.43}$$

using $(\mathbf{p}\cdot\boldsymbol{\sigma})\sigma^i(\mathbf{p}\cdot\boldsymbol{\sigma})=-\sigma^i(\mathbf{p}\cdot\boldsymbol{\sigma})^2+2p^i(\mathbf{p}\cdot\boldsymbol{\sigma})$ and $\psi_r^\dagger\sigma^k\psi_s=\sigma^k_{r,s}$.

The first identity can be used to simplify j^μ_{reg}, and we find indeed using our previous calculation (34) of the norm $\mathcal{N}$,

$$j^i_{\text{reg}} = \int \frac{d^3\mathbf{p}}{(2\pi)^3} 2p^i \sum_{r=1}^{2} \left[|b_r(\mathbf{p})|^2 + |d_r(\mathbf{p})|^2\right] = (4\pi d^2)^{3/2} \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{p^i}{p^0} e^{-\mathbf{p}^2 d^2} \tag{1.45}$$

By a parity argument, $\mathbf{p} \to -\mathbf{p}$, we have

$$j^i_{\text{reg}} = (4\pi d^2)^{3/2} \frac{1}{2} \left(\int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{p^i}{p^0} e^{-\mathbf{p}^2 d^2} - \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{p^i}{p^0} e^{-\mathbf{p}^2 d^2} \right) = 0. \tag{1.46}$$

Next, we turn to $\mathbf{j}_{\text{osc}}$, and we need to show that it is actually vanishing. We compute

$$\begin{aligned} \sum_{r,s} b^*_r(\mathbf{p}) \sigma^k_{r,s} d_s(-\mathbf{p}) &= -F(\mathbf{p})^2 (p^0 + m) \xi^\dagger \sigma^k (\mathbf{p} \cdot \boldsymbol{\sigma}) \xi = -F(\mathbf{p})^2 (p^0 + m) p^l \xi^\dagger \sigma^k \sigma^l \xi, \\ \sum_{r,s} d^*_r(\mathbf{p}) \sigma^k_{r,s} b_s(-\mathbf{p}) &= F(\mathbf{p})^2 (p^0 + m) \xi^\dagger (\mathbf{p} \cdot \boldsymbol{\sigma}) \sigma^k \xi = F(\mathbf{p})^2 (p^0 + m) p^l \xi^\dagger \sigma^l \sigma^k \xi. \end{aligned} \tag{1.47}$$

Putting everything together, we find

$$j^i_{\text{osc}} = \int \frac{d^3\mathbf{p}}{(2\pi)^3} F(\mathbf{p})^2 \left(2p^0 (p^0 + m) \delta^{ik} p^l - 2p^i p^k p^l \right) \xi^\dagger \left(-e^{2itp^0} \sigma^k \sigma^l + e^{-2itp^0} \sigma^l \sigma^k \right) \xi \tag{1.48}$$

Then, we use

$$-e^{2itp^0} \sigma^k \sigma^l + e^{-2itp^0} \sigma^l \sigma^k = -[\sigma^k, \sigma^l] \cos(2tp^0) - i \sin(2tp^0) \{\sigma^k, \sigma^l\} = -2i\varepsilon^{klm} \sigma^m \cos(2tp^0) - 2i\delta^{kl} \sin(2tp^0). \tag{1.49}$$

to rewrite (exploiting the symmetry)

$$\begin{aligned} j^i_{\text{osc}} &= -4i \int \frac{d^3\mathbf{p}}{(2\pi)^3} F(\mathbf{p})^2 \left(p^0 (p^0 + m) (\cos(2tp^0) \varepsilon^{ijk} p^j \xi^\dagger \sigma^k \xi - p^i \sin(2tp^0)) - p^i \mathbf{p}^2 \sin(2tp^0) \right) \\ &= 4i \int \frac{d^3\mathbf{p}}{(2\pi)^3} F(\mathbf{p})^2 \left(m(p^0 + m) p^i \sin(2tp^0) - p^0 (p^0 + m) \cos(2tp^0) \varepsilon^{ijk} p^j \xi^\dagger \sigma^k \xi \right). \end{aligned} \tag{1.50}$$

Finally, we compute

$$\xi^\dagger \sigma^1 \xi = 1, \quad \xi^\dagger \sigma^2 \xi = 0, \quad \xi^\dagger \sigma^3 \xi = 0 \quad \Rightarrow \quad \varepsilon^{ijk} p^j \xi^\dagger \sigma^k \xi = \mathbf{p} \wedge \mathbf{e}_1. \tag{1.51}$$

It gives

$$\mathbf{j}_{\text{osc}} = i(4\pi d^2)^{3/2} \int \frac{d^3\mathbf{p}}{(2\pi)^3} e^{-\mathbf{p}^2 d^2} \left[\frac{m\mathbf{p}}{(p^0)^2} \sin(2tp^0) - \frac{\mathbf{p}}{p^0} \wedge \mathbf{e}_1 \cos(2tp^0) \right], \quad \mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}. \tag{1.52}$$

At this stage, it seems puzzling because the current appears to be purely imaginary. But observe that the integrand $G(\mathbf{p})$ satisfies the symmetry $G(-\mathbf{p}) = -G(\mathbf{p})$, and so

$$\mathbf{j}_{\text{osc}} = \int_{-\infty}^{\infty} G(\mathbf{p}) d^3\mathbf{p} = -\int_{-\infty}^{\infty} G(\mathbf{p}) d^3\mathbf{p} = 0. \tag{1.53}$$

Of course, we could have used this argument much earlier, but I believe the full calculation is instructive :

For a general wave packet, these two terms of the current $\mathbf{j}_{\text{reg}}$ and $\mathbf{j}_{\text{osc}}$ are non-vanishing. The term $\mathbf{j}_{\text{osc}}$, arising due to the presence of negative energy solutions, is at the origin of a phenomenon of oscillations called Zitterbewegung and first observed by Schrödinger.

1.3 Assignment 2

Exercise 1. *(10 points) In this exercise, we consider a massive real scalar field, with the mode decomposition*

$$\Phi(x) = \frac{1}{(2\pi)^{3/2}} \int \frac{d^3\mathbf{k}}{2k^0} (a_{\mathbf{k}} e^{-ikx} + a_{\mathbf{k}}^\dagger e^{ikx}), \quad [a_{\mathbf{k}}, a_{\mathbf{k}'}^\dagger] = 2k^0 \delta^{(3)}(\mathbf{k} - \mathbf{k}'). \tag{1.54}$$

and $[a_{\mathbf{k}}, a_{\mathbf{k}'}] = [a_{\mathbf{k}}^\dagger, a_{\mathbf{k}'}^\dagger] = 0$*. We also recall the expression of the quantum Hamiltonian, and the corresponding evolution operators*

$$H = \frac{1}{2} \int d^3\mathbf{k}\, a_{\mathbf{k}}^\dagger a_{\mathbf{k}}, \quad U(t) = e^{-iHt} \tag{1.55}$$

1. Compute the adjoint action of the evolution operator on the modes $a_{\mathbf{k}}$ and $a_{\mathbf{k}}^\dagger$, and deduce that the evolution of the quantum field is indeed given by *(2 points)*

$$\Phi(t, \mathbf{x}) = U(t)^\dagger \Phi(0, \mathbf{x}) U(t). \tag{1.56}$$

2. We are interested in the family of coherent states associated to L^2 functions $f(\mathbf{k}) : \mathbb{R}^3 \to \mathbb{C}$, and defined as

$$|f\rangle = \exp\left(\int \frac{d^3\mathbf{k}}{2k^0} f(\mathbf{k}) a_{\mathbf{k}}^\dagger \right) |0\rangle. \tag{1.57}$$

   Compute the inner product $\langle f_1 | f_2 \rangle$ of two coherent states associated to different functions f_1 and f_2. Are these states orthogonal? *(3 points)*
3. Show that $a_{\mathbf{k}}$ is diagonal on the state $|f\rangle$. What is the corresponding eigenvalue? *(3 points)*
4. Show that coherent states remain coherent under time evolution, and more specifically *(2 points)*

$$U(t)|f\rangle = |f_t\rangle, \quad \text{with} \quad f_t(\mathbf{k}) = e^{-it\sqrt{\mathbf{k}^2 + m^2}} f(\mathbf{k}). \tag{1.58}$$

Solutions:

1. We use the formula for two operators A and B,

$$e^A B e^{-A} = e^{\mathrm{ad}_A} B = \sum_{n=0}^{\infty} \frac{(\mathrm{ad}_A)^n}{n!} B, \quad \mathrm{ad}_A B = [A, B]. \tag{1.59}$$

We have

$$\begin{aligned} [H, a_{\mathbf{k}}] &= \frac{1}{2} \int \frac{d^3\mathbf{k}'}{2k'^0} [a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'}, a_{\mathbf{k}}] = -\frac{1}{2} \int \frac{d^3\mathbf{k}'}{2k'^0} 2k^0 \delta^{(3)}(\mathbf{k} - \mathbf{k}') a_{\mathbf{k}'} = -k^0 a_{\mathbf{k}}, \\ [H, a_{\mathbf{k}}^\dagger] &= \frac{1}{2} \int \frac{d^3\mathbf{k}'}{2k'^0} [a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'}, a_{\mathbf{k}}^\dagger] = \frac{1}{2} \int \frac{d^3\mathbf{k}'}{2k'^0} 2k^0 \delta^{(3)}(\mathbf{k} - \mathbf{k}') a_{\mathbf{k}'}^\dagger = k^0 a_{\mathbf{k}}^\dagger \end{aligned} \tag{1.60}$$

and so

$$\begin{aligned} &[itH, a_{\mathbf{k}}] = -itk^0 a_{\mathbf{k}} \quad \Rightarrow \quad (\mathrm{ad}_{itH})^n a_{\mathbf{k}} = (-itk^0)^n a_{\mathbf{k}} \quad \Rightarrow \quad U(t)^\dagger a_{\mathbf{k}} U(t) = e^{itH} a_{\mathbf{k}} e^{-itH} = e^{-itk^0} a_{\mathbf{k}}, \\ &[itH, a_{\mathbf{k}}^\dagger] = itk^0 a_{\mathbf{k}} \quad \Rightarrow \quad (\mathrm{ad}_{itH})^n a_{\mathbf{k}}^\dagger = (itk^0)^n a_{\mathbf{k}}^\dagger \quad \Rightarrow \quad U(t)^\dagger a_{\mathbf{k}}^\dagger U(t) = e^{itH} a_{\mathbf{k}}^\dagger e^{-itH} = e^{itk^0} a_{\mathbf{k}}^\dagger, \end{aligned} \tag{1.61}$$

since $H^\dagger = H$ and so $U(t)^\dagger = U(t)^{-1} = U(-t)$. Thus, we find

$$\begin{aligned} U(t)^\dagger \Phi(0,\mathbf{x}) U(t) &= \frac{1}{(2\pi)^{3/2}} \int \frac{d^3\mathbf{k}}{2k^0} U(t)^\dagger (a_\mathbf{k} e^{-i\mathbf{k}\cdot\mathbf{x}} + a_\mathbf{k}^\dagger e^{i\mathbf{k}\cdot\mathbf{x}}) U(t) \\ &= \frac{1}{(2\pi)^{3/2}} \int \frac{d^3\mathbf{k}}{2k^0} (U(t)^\dagger a_\mathbf{k} U(t) e^{i\mathbf{k}\cdot\mathbf{x}} + U(t)^\dagger a_\mathbf{k}^\dagger U(t) e^{-i\mathbf{k}\cdot\mathbf{x}}) \\ &= \frac{1}{(2\pi)^{3/2}} \int \frac{d^3\mathbf{k}}{2k^0} (e^{-ik^0 t} a_\mathbf{k} e^{i\mathbf{k}\cdot\mathbf{x}} + e^{ik^0 t} a_\mathbf{k}^\dagger e^{-i\mathbf{k}\cdot\mathbf{x}}) \\ &= \frac{1}{(2\pi)^{3/2}} \int \frac{d^3\mathbf{k}}{2k^0} (e^{-ikx} a_\mathbf{k} + a_\mathbf{k}^\dagger e^{ikx}) \\ &= \Phi(t,\mathbf{x}). \end{aligned} \tag{1.62}$$

2. Expanding the exponentials, we have

$$\langle f_1 | f_2 \rangle = \sum_{n,m=0}^{\infty} \frac{1}{n!m!} \int \prod_{\alpha=1}^{n} \left[f_1^*(\mathbf{k}_\alpha) \frac{d^3\mathbf{k}_\alpha}{2k_\alpha^0} \right] \prod_{\beta=1}^{m} \left[f_2(\mathbf{k}'_\beta) \frac{d^3\mathbf{k}'_\beta}{2k'^0_\beta} \right] \langle 0 | a_{\mathbf{k}_1} \cdots a_{\mathbf{k}_n} a_{\mathbf{k}'_1}^\dagger \cdots a_{\mathbf{k}'_m}^\dagger | 0 \rangle. \tag{1.63}$$

Let

$$C_{n,m} = \langle 0 | a_{\mathbf{k}_1} \cdots a_{\mathbf{k}_n} a_{\mathbf{k}'_1}^\dagger \cdots a_{\mathbf{k}'_m}^\dagger | 0 \rangle. \tag{1.64}$$

By induction, one can show that

$$C_{n,m}(\mathbf{k}_1, \cdots, \mathbf{k}_n | \mathbf{k}'_1, \cdots, \mathbf{k}'_m) = \delta_{n,m} \sum_{\sigma \in S_n} \prod_{\alpha=1}^{n} 2k_\alpha^0 \delta^{(3)}(\mathbf{k}_\alpha - \mathbf{k}'_{\sigma(\alpha)}). \tag{1.65}$$

Let's first consider the case $n = m$. We obviously have $C_{00} = 1$, and using the commutation relations, we find $C_{11} = 2k_1^0 \delta^{(3)}(\mathbf{k}_1 - \mathbf{k}'_1)$. The first non-trivial case is

$$\begin{aligned} C_{22}(\mathbf{k}_1, \mathbf{k}_2 | \mathbf{k}'_1, \mathbf{k}'_2) &= \langle 0 | a_{\mathbf{k}_1} a_{\mathbf{k}_2} a_{\mathbf{k}'_1}^\dagger a_{\mathbf{k}'_2}^\dagger | 0 \rangle \\ &= 2k_2^0 \delta^{(3)}(\mathbf{k}_2 - \mathbf{k}'_1) \langle 0 | a_{\mathbf{k}_1} a_{\mathbf{k}'_2}^\dagger | 0 \rangle + 2k_2^0 \delta^{(3)}(\mathbf{k}_2 - \mathbf{k}'_2) \langle 0 | a_{\mathbf{k}_1} a_{\mathbf{k}'_1}^\dagger | 0 \rangle \\ &= 2k_2^0 \delta^{(3)}(\mathbf{k}_2 - \mathbf{k}'_1) C_{11}(\mathbf{k}_1 | \mathbf{k}'_2) + 2k_2^0 \delta^{(3)}(\mathbf{k}_2 - \mathbf{k}'_2) C_{11}(\mathbf{k}_1 | \mathbf{k}'_1) \end{aligned} \tag{1.66}$$

where the second line is obtained by moving $a_{\mathbf{k}_2}$ to the right using the commutator until it reaches the state $|0\rangle$ where the corresponding term vanishes since $a_{\mathbf{k}_2}|0\rangle = 0$. Repeating this argument, we have

$$C_{nn}(\mathbf{k}_1, \cdots, \mathbf{k}_n | \mathbf{k}'_1, \cdots, \mathbf{k}'_n) = \sum_{\beta=1}^{n} 2k_n^{(0)} \delta^{(3)}(\mathbf{k}_n - \mathbf{k}_\beta) C_{n-1\ n-1}(\mathbf{k}_1, \cdots, \mathbf{k}_{n-1} | \mathbf{k}'_1, \cdots, \cancel{\mathbf{k}'_\alpha}, \cdots, \mathbf{k}'_n). \tag{1.67}$$

Thus, by induction we have

$$\begin{aligned} C_{nn}(\mathbf{k}_1, \cdots, \mathbf{k}_n | \mathbf{k}'_1, \cdots, \mathbf{k}'_n) &= \sum_{\beta=1}^{n} 2k_n^{(0)} \delta^{(3)}(\mathbf{k}_n - \mathbf{k}_\beta) \sum_{\sigma \in S_{n-1}} \prod_{\alpha=1}^{n-1} 2k_\alpha^0 \delta^{(3)}(\mathbf{k}_\alpha - \mathbf{k}'_{\sigma(\alpha)}) \\ &= \sum_{\sigma \in S_n} \prod_{\alpha=1}^{n} 2k_\alpha^0 \delta^{(3)}(\mathbf{k}_\alpha - \mathbf{k}'_{\sigma(\alpha)}) \end{aligned} \tag{1.68}$$

denoting $\beta = \sigma(n)$. It remains to show the orthogonality $C_{n,m} \propto \delta_{n,m}$, this is done by noticing that if $n > m$ or $n < m$ the commutation procedure used above will terminate with extra insertions of a or $a^\dagger$,

and so the corresponding correlator should vanish. For instance,

$$C_{21}(\mathbf{k}_1,\mathbf{k}_2|\mathbf{k}'_1) = \langle 0|a_{\mathbf{k}_1}a_{\mathbf{k}_2}a^\dagger_{\mathbf{k}'_1}|0\rangle = 2k_2^0\delta^{(3)}(\mathbf{k}_2-\mathbf{k}'_1)\langle 0|a_{\mathbf{k}_1}|0\rangle = 0,$$
$$C_{12}(\mathbf{k}_1|\mathbf{k}'_1,\mathbf{k}'_2) = \langle 0|a_{\mathbf{k}_1}a^\dagger_{\mathbf{k}'_1}a^\dagger_{\mathbf{k}'_2}|0\rangle = 2k_1^0\delta^{(3)}(\mathbf{k}_1-\mathbf{k}'_1)\langle 0|a^\dagger_{\mathbf{k}'_2}|0\rangle + 2k_1^0\delta^{(3)}(\mathbf{k}_1-\mathbf{k}'_2)\langle 0|a^\dagger_{\mathbf{k}'_1}|0\rangle = 0. \tag{1.69}$$

Coming back to the expression of the inner product, and inserting our result for C_{nn}, we find

$$\begin{aligned}\langle f_1|f_2\rangle &= \sum_{n,m=0}^{\infty}\frac{1}{n!m!}\int\prod_{\alpha=1}^{n}\left[f_1^*(\mathbf{k}_\alpha)\frac{d^3\mathbf{k}_\alpha}{2k_\alpha^0}\right]\prod_{\beta=1}^{m}\left[f_2(\mathbf{k}'_\beta)\frac{d^3\mathbf{k}'_\beta}{2k'^0_\beta}\right]\delta_{n,m}\sum_{\sigma\in S_n}\prod_{\alpha=1}^{n}2k_\alpha^0\delta^{(3)}(\mathbf{k}_\alpha-\mathbf{k}'_{\sigma(\alpha)})\\
&= \sum_{n=0}^{\infty}\frac{1}{(n!)^2}\sum_{\sigma\in S_n}\int\prod_{\alpha=1}^{n}\left[f_1^*(\mathbf{k}_\alpha)\frac{d^3\mathbf{k}_\alpha}{2k_\alpha^0}\right]\prod_{\beta=1}^{n}\left[f_2(\mathbf{k}'_\beta)\frac{d^3\mathbf{k}'_\beta}{2k'^0_\beta}\right]\prod_{\alpha=1}^{n}\left[2k_\alpha^0\delta^{(3)}(\mathbf{k}_\alpha-\mathbf{k}'_{\sigma(\alpha)})\right]\\
&= \sum_{n=0}^{\infty}\frac{1}{n!}\int\prod_{\alpha=1}^{n}\left[f_1^*(\mathbf{k}_\alpha)\frac{d^3\mathbf{k}_\alpha}{2k_\alpha^0}\right]\prod_{\beta=1}^{n}\left[f_2(\mathbf{k}'_\beta)\frac{d^3\mathbf{k}'_\beta}{2k'^0_\beta}\right]\delta_{n,m}\prod_{\alpha=1}^{n}\left[2k_\alpha^0\delta^{(3)}(\mathbf{k}_\alpha-\mathbf{k}'_\alpha)\right]\\
&= \sum_{n=0}^{\infty}\frac{1}{n!}\int\prod_{\alpha=1}^{n}\left[\frac{d^3\mathbf{k}_\alpha}{2k_\alpha^0}f_1^*(\mathbf{k}_\alpha)f_2(\mathbf{k}_\alpha)\right] = \exp\left(\int\frac{d^3\mathbf{k}}{2k^0}f_1^*(\mathbf{k})f_2(\mathbf{k})\right)\end{aligned} \tag{1.70}$$

by exploiting the invariance under permutation of the integral. In general, the integral inside the exponential is non-vanishing and the states are not orthogonal.

3. Let's first compute by induction

$$[a_{\mathbf{k}}, a^\dagger_{\mathbf{k}_1}\cdots a^\dagger_{\mathbf{k}_n}] = \sum_{\alpha=1}^{n}2k_\alpha^{(0)}\delta^{(3)}(\mathbf{k}-\mathbf{k}_\alpha)a^\dagger_{\mathbf{k}_1}\cdots\cancel{a^\dagger_{\mathbf{k}_\alpha}}\cdots a^\dagger_{\mathbf{k}_n}, \tag{1.71}$$

and since $a_{\mathbf{k}}|0\rangle = 0$, we deduce

$$\begin{aligned}a_{\mathbf{k}}\int\prod_{\alpha=1}^{n}\left[\frac{d^3\mathbf{k}_\alpha}{2k_\alpha^0}f(\mathbf{k}_\alpha)\right]a^\dagger_{\mathbf{k}_1}\cdots a^\dagger_{\mathbf{k}_n}|0\rangle &= \sum_{\beta=1}^{n}\int\prod_{\alpha=1}^{n}\left[\frac{d^3\mathbf{k}_\alpha}{2k_\alpha^0}f(\mathbf{k}_\alpha)\right]2k_\beta^{(0)}\delta^{(3)}(\mathbf{k}-\mathbf{k}_\beta)a^\dagger_{\mathbf{k}_1}\cdots\cancel{a^\dagger_{\mathbf{k}_\beta}}\cdots a^\dagger_{\mathbf{k}_n}|0\rangle\\
&= n\int\prod_{\alpha=1}^{n}\left[\frac{d^3\mathbf{k}_\alpha}{2k_\alpha^0}f(\mathbf{k}_\alpha)\right]2k_n^{(0)}\delta^{(3)}(\mathbf{k}-\mathbf{k}_n)a^\dagger_{\mathbf{k}_1}\cdots a^\dagger_{\mathbf{k}_{n-1}}|0\rangle\\
&= nf(\mathbf{k})\int\prod_{\alpha=1}^{n-1}\left[\frac{d^3\mathbf{k}_\alpha}{2k_\alpha^0}f(\mathbf{k}_\alpha)\right]a^\dagger_{\mathbf{k}_1}\cdots a^\dagger_{\mathbf{k}_{n-1}}|0\rangle,\end{aligned} \tag{1.72}$$

since we can always commute the operators and relabel the variables to take $\beta = n$. We deduce that

$$\begin{aligned}a_{\mathbf{k}}|f\rangle &= \sum_{n=0}^{\infty}\frac{1}{n!}a_{\mathbf{k}}\int\prod_{\alpha=1}^{n}\left[\frac{d^3\mathbf{k}_\alpha}{2k_\alpha^0}f(\mathbf{k}_\alpha)\right]a^\dagger_{\mathbf{k}_1}\cdots a^\dagger_{\mathbf{k}_n}|0\rangle\\
&= \sum_{n=1}^{\infty}\frac{1}{(n-1)!}f(\mathbf{k})\int\prod_{\alpha=1}^{n-1}\left[\frac{d^3\mathbf{k}_\alpha}{2k_\alpha^0}f(\mathbf{k}_\alpha)\right]a^\dagger_{\mathbf{k}_1}\cdots a^\dagger_{\mathbf{k}_{n-1}}|0\rangle\\
&= f(\mathbf{k})|f\rangle.\end{aligned} \tag{1.73}$$

Thus $a_{\mathbf{k}}$ is diagonal on the states $|f\rangle$ with eigenvalue $f(\mathbf{k})$.

4. This calculation is very similar to the case of the field solved in question one. Using the result on

the time evolution of the modes, we can write down

$$\begin{aligned}
U(t)|f\rangle &= \sum_{n=0}^{\infty}\frac{1}{n!}\int\prod_{\alpha=1}^{n}\left[\frac{d^3\mathbf{k}_\alpha}{2k_\alpha^0}f(\mathbf{k}_\alpha)\right]U(t)a^\dagger_{\mathbf{k}_1}\cdots a^\dagger_{\mathbf{k}_n}|0\rangle \\
&= \sum_{n=0}^{\infty}\frac{1}{n!}\int\prod_{\alpha=1}^{n}\left[\frac{d^3\mathbf{k}_\alpha}{2k_\alpha^0}f(\mathbf{k}_\alpha)\right]U(t)a^\dagger_{\mathbf{k}_1}U(t)^{-1}U(t)a^\dagger_{\mathbf{k}_2}U(t)^{-1}\cdots U(t)a^\dagger_{\mathbf{k}_n}U(t)^{-1}U(t)|0\rangle \\
&= \sum_{n=0}^{\infty}\frac{1}{n!}\int\prod_{\alpha=1}^{n}\left[\frac{d^3\mathbf{k}_\alpha}{2k_\alpha^0}f(\mathbf{k}_\alpha)\right]e^{-ik_1^0t}a^\dagger_{\mathbf{k}_1}e^{-ik_2^0t}a^\dagger_{\mathbf{k}_2}\cdots e^{-ik_n^0t}a^\dagger_{\mathbf{k}_n}|0\rangle \\
&= \sum_{n=0}^{\infty}\frac{1}{n!}\int\prod_{\alpha=1}^{n}\left[\frac{d^3\mathbf{k}_\alpha}{2k_\alpha^0}e^{-ik_\alpha^0t}f(\mathbf{k}_\alpha)\right]a^\dagger_{\mathbf{k}_1}a^\dagger_{\mathbf{k}_2}\cdots a^\dagger_{\mathbf{k}_n}|0\rangle = |f_t\rangle,
\end{aligned} \tag{1.74}$$

where we used $U(t)|0\rangle = |0\rangle$ since $H|0\rangle = 0$.

Exercise 2. *(10 points) In this exercise, we study both the classical and quantum Dirac spinor fields of mass m.*

1. Let $\boldsymbol{\sigma} = (\sigma^1, \sigma^2, \sigma^3)$ be the vector of Pauli matrices and $\mathbf{k} = (k^1, k^2, k^3)$ the momentum vector. Find two orthonormal eigenvectors of the matrix $\mathbf{k}\cdot\boldsymbol{\sigma}$ with eigenvalues $|\mathbf{k}|$ and $-|\mathbf{k}|$. In the following, we will denote them $\varphi_1(\mathbf{k})$ and $\varphi_2(\mathbf{k})$ respectively. *(1 point)*
2. We introduce the following 4-vector of polarization,

$$U_r(\mathbf{k}) = \frac{1}{\sqrt{k^0+m}}\begin{pmatrix}(k^0+m)\varphi_r(\mathbf{k})\\ \mathbf{k}\cdot\boldsymbol{\sigma}\varphi_r(\mathbf{k})\end{pmatrix}, \quad V_r(\mathbf{k}) = \frac{1}{\sqrt{k^0+m}}\begin{pmatrix}\mathbf{k}\cdot\boldsymbol{\sigma}\varphi_r(\mathbf{k})\\ (k^0+m)\varphi_r(\mathbf{k})\end{pmatrix}, \tag{1.75}$$

   and the classical field

$$\Psi(x) = \frac{1}{(2\pi)^{3/2}}\int\frac{d^3\mathbf{k}}{2k^0}\sum_{r=1,2}(U_r(\mathbf{k})b_{r\mathbf{k}}e^{-ikx} + V_r(\mathbf{k})d^*_{r\mathbf{k}}e^{ikx}), \tag{1.76}$$

   with arbitrary modes $b_{r\mathbf{k}}$ and $d^*_{r\mathbf{k}}$ determined from initial conditions. Explain why $\Psi(x)$ satisfies the Dirac equation. *(1 point)*
3. Show that the following projection relations hold *(1 point)*

$$U_r^\dagger(\mathbf{k})U_s(\mathbf{k}) = V_r^\dagger(\mathbf{k})V_s(\mathbf{k}) = 2k^0\delta_{r,s}, \quad U_r^\dagger(\mathbf{k})V_s(-\mathbf{k}) = V_r^\dagger(\mathbf{k})U_s(-\mathbf{k}) = 0. \tag{1.77}$$

4. We define the helicity operator $S_\mathbf{k} = \frac{\mathbf{k}\cdot\boldsymbol{\Sigma}}{|\mathbf{k}|}$ where $\boldsymbol{\Sigma}$ is the vector of 4×4 matrices $\boldsymbol{\Sigma} = \begin{pmatrix}\boldsymbol{\sigma} & 0\\ 0 & \boldsymbol{\sigma}\end{pmatrix}$. Show that $U_r(\mathbf{k})$ and $V_r(\mathbf{k})$ are eigenvectors of $S_\mathbf{k}$ with eigenvalues $(-1)^{r+1}$. *(1 point)*
5. We now introduce a quantum Dirac field with the same polarization vectors,

$$\Psi(x) = \frac{1}{(2\pi)^{3/2}}\int\frac{d^3\mathbf{k}}{2k^0}\sum_{r=1,2}(U_r(\mathbf{k})b_{r\mathbf{k}}e^{-ikx} + V_r(\mathbf{k})d^\dagger_{r\mathbf{k}}e^{ikx}), \tag{1.78}$$

   with the modes satisfying the anticommutation relations $\{b_{r\mathbf{k}}, b^\dagger_{s\mathbf{k}'}\} = \{d_{r\mathbf{k}}, d^\dagger_{s\mathbf{k}'}\} = 2k^0\delta^{(3)}(\mathbf{k}-\mathbf{k}')\delta_{r,s}$ (other anticommutators being trivial). The quantum helicity operator can be obtained as a projection of the Pauli-Lubanski operator on the direction of momentum. It leads to the following expression

$$\hat{S} = \frac{-i}{2}\int d^3\mathbf{x} : \Psi^\dagger(x)\boldsymbol{\Sigma}\cdot\hat{\nabla}\Psi(x) :, \tag{1.79}$$

where the differential operator $\hat{\nabla}$ acts as follows on the Dirac field,

$$\hat{\nabla}\Psi(x) = \frac{i}{(2\pi)^{3/2}} \int \frac{d^3\mathbf{k}}{2k^0} \frac{\mathbf{k}}{|\mathbf{k}|} \sum_{r=1,2} (U_r(\mathbf{k}) b_{r\mathbf{k}} e^{-ikx} - V_r(\mathbf{k}) d^\dagger_{r\mathbf{k}} e^{ikx}) \tag{1.80}$$

Show that in our polarization basis, the helicity operator simplifies into *(3 points)*

$$\hat{S} = \frac{1}{2} \int \frac{d^3\mathbf{k}}{2k^0} \sum_{r=1,2} (-)^{r+1} (b^\dagger_{r\mathbf{k}} b_{r\mathbf{k}} + d^\dagger_{r\mathbf{k}} d_{r\mathbf{k}}). \tag{1.81}$$

6. We consider a state with one particle and one antiparticle with the same momentum and independent polarizations,

$$|1\bar{1}rs\rangle = b^\dagger_{r\mathbf{k}} d^\dagger_{s\mathbf{k}} |0\rangle. \tag{1.82}$$

Compute the energy, momentum, helicity and U(1) charge of this state. *(3 points)*

Solutions:

1. Solutions might differ by a phase, but using your favorite method you should find

$$\varphi_1(\mathbf{k}) = \frac{1}{\sqrt{2|\mathbf{k}|(|\mathbf{k}| - k^3)}} \begin{pmatrix} k^1 - ik^2 \\ -k^3 + |\mathbf{k}| \end{pmatrix}, \quad \varphi_2(\mathbf{k}) = \frac{1}{\sqrt{2|\mathbf{k}|(|\mathbf{k}| + k^3)}} \begin{pmatrix} k^1 - ik^2 \\ -k^3 - |\mathbf{k}| \end{pmatrix}. \tag{1.83}$$

Note that we have indeed $\varphi_r^\dagger \varphi_s = \delta_{r,s}$.

2. Using the Fourier transform, we have shown in class that solutions of the Dirac equations are of the form (23) where polarization vectors $U_r(\mathbf{k})$ and $V_r(\mathbf{k})$ must satisfy the equations

$$(\not{k} - m) U_r(\mathbf{k}) = 0, \quad (\not{k} + m) V_r(\mathbf{k}) = 0. \tag{1.84}$$

Then, we have shown that the solutions of these equations are given by vectors of the form

$$U_r(\mathbf{k}) = \frac{1}{\sqrt{k^0 + m}} \begin{pmatrix} (k^0 + m)\psi_r \\ \mathbf{k} \cdot \boldsymbol{\sigma} \psi_r \end{pmatrix}, \quad V_r(\mathbf{k}) = \frac{1}{\sqrt{k^0 + m}} \begin{pmatrix} \mathbf{k} \cdot \boldsymbol{\sigma} \chi_r \\ (k^0 + m)\chi_r \end{pmatrix}. \tag{1.85}$$

where ψ_r (and χ_r) are two arbitrary linearly independent two dimensional vectors. The choice $\psi_r = \chi_r = \varphi_r(\mathbf{k})$ is just a particular case.

3. We use the same method as seen in class, exploiting the fact that $\varphi_r^\dagger \varphi_s = \delta_{r,s}$, $\boldsymbol{\sigma}^\dagger = \boldsymbol{\sigma}$ and the mass shell condition $m^2 = (k^0)^2 - \mathbf{k}^2$. For instance,

$$\begin{aligned} U_r(\mathbf{k})^\dagger U_s(\mathbf{k}) &= \frac{1}{k^0 + m} \left[\varphi_r(\mathbf{k})^\dagger (k^0 + m)^2 \varphi_s(\mathbf{k}) + \varphi_r(\mathbf{k})^\dagger (\mathbf{k} \cdot \boldsymbol{\sigma})^2 \varphi_s(\mathbf{k}) \right] \\ &= \frac{1}{k^0 + m} \left[(k^0 + m)^2 + \mathbf{k}^2 \right] \varphi_r(\mathbf{k})^\dagger \varphi_s(\mathbf{k}) = 2k^0 \delta_{r,s}, \end{aligned} \tag{1.86}$$

since $(\mathbf{k} \cdot \boldsymbol{\sigma})^2 = k^i k^j \sigma^i \sigma^j = k^i k^j \delta_{i,j} = \mathbf{k}^2$, and

$$U_r(\mathbf{k})^\dagger V_s(-\mathbf{k}) = \frac{1}{k^0 + m} \left[-\varphi_r(\mathbf{k})^\dagger (k^0 + m)(\mathbf{k} \cdot \boldsymbol{\sigma}) \varphi_s(\mathbf{k}) + \varphi_r(\mathbf{k})^\dagger (\mathbf{k} \cdot \boldsymbol{\sigma})(k^0 + m) \varphi_s(\mathbf{k}) \right] = 0. \tag{1.87}$$

The other two relations are obtained in the same way.

4. Using the property $(\mathbf{k}\cdot\boldsymbol{\sigma})\varphi_r(\mathbf{k}) = (-)^{r+1}|\mathbf{k}|$ and $(\mathbf{k}\cdot\boldsymbol{\sigma})^2 = \mathbf{k}^2$, we compute

$$\begin{aligned}(\mathbf{k}\cdot\boldsymbol{\Sigma})U_r(\mathbf{k}) &= \frac{1}{\sqrt{k^0+m}}\begin{pmatrix}\mathbf{k}\cdot\boldsymbol{\sigma} & 0\\ 0 & \mathbf{k}\cdot\boldsymbol{\sigma}\end{pmatrix}\begin{pmatrix}(k^0+m)\varphi_r(\mathbf{k})\\ \mathbf{k}\cdot\boldsymbol{\sigma}\varphi_r(\mathbf{k})\end{pmatrix} = \frac{1}{\sqrt{k^0+m}}\begin{pmatrix}(k^0+m)\mathbf{k}\cdot\boldsymbol{\sigma}\varphi_r(\mathbf{k})\\ \mathbf{k}^2\varphi_r(\mathbf{k})\end{pmatrix}\\ &= \frac{(-)^{r+1}|\mathbf{k}|}{\sqrt{k^0+m}}\begin{pmatrix}(k^0+m)\varphi_r(\mathbf{k})\\ (\mathbf{k}\cdot\boldsymbol{\sigma})\varphi_r(\mathbf{k})\end{pmatrix} = (-)^{r+1}|\mathbf{k}|U_r(\mathbf{k}) \quad\Rightarrow\quad S_{\mathbf{k}}U_r(\mathbf{k}) = (-)^{r+1}U_r(\mathbf{k}).\end{aligned} \tag{1.88}$$

The calculation is exactly the same for $V_r(\mathbf{k})$.

5. This calculation is very similar to what we have been doing in class, so we will be brief. From the expression of the operators $\hat{S}$ and $\hat{\nabla}$, we find

$$\hat{S} = \frac{1}{2(2\pi)^3}\int d^3\mathbf{x}\int\frac{d^3\mathbf{k}}{2k^0}\int\frac{d^3\mathbf{k}'}{2k'^0}\sum_{r,s=1,2}:\left(U_r(\mathbf{k})^\dagger b^\dagger_{r\mathbf{k}}e^{ikx} + V_r(\mathbf{k})^\dagger d_{r\mathbf{k}}e^{-ikx}\right)\frac{\mathbf{k}'\cdot\boldsymbol{\Sigma}}{|\mathbf{k}'|}\left(U_s(\mathbf{k}')b_{s\mathbf{k}'}e^{-ik'x} - V_s(\mathbf{k}')d^\dagger_{s\mathbf{k}'}e^{i} \tag{1.89}$$

We start by performing the $d^3\mathbf{x}$ integration, it produces delta functions (see Assignment 1),

$$\begin{aligned}\hat{S} = \frac{1}{2}\int\frac{d^3\mathbf{k}}{2k^0}\int\frac{d^3\mathbf{k}'}{2k'^0}\sum_{r,s=1,2}\Bigg[&\delta^{(3)}(\mathbf{k}-\mathbf{k}')\left(U_r(\mathbf{k})^\dagger\frac{\mathbf{k}'\cdot\boldsymbol{\Sigma}}{|\mathbf{k}'|}U_s(\mathbf{k}'):b^\dagger_{r\mathbf{k}}b_{s\mathbf{k}'}: - V_r(\mathbf{k})^\dagger\frac{\mathbf{k}'\cdot\boldsymbol{\Sigma}}{|\mathbf{k}'|}V_s(\mathbf{k}'):d_{r\mathbf{k}}d^\dagger_{s\mathbf{k}'}:\right)\\ &+\delta^{(3)}(\mathbf{k}+\mathbf{k}')\left(-e^{2ik^0t}U_r(\mathbf{k})^\dagger\frac{\mathbf{k}'\cdot\boldsymbol{\Sigma}}{|\mathbf{k}'|}V_s(\mathbf{k}'):b^\dagger_{r\mathbf{k}}d^\dagger_{s\mathbf{k}'}: + e^{-2ik^0t}V_r(\mathbf{k})^\dagger\frac{\mathbf{k}'\cdot\boldsymbol{\Sigma}}{|\mathbf{k}'|}U_s(\mathbf{k}'):d_r\end{aligned} \tag{1.90}$$

Then, we use the fact that polarization vectors are eigenvectors of the operator $S_{\mathbf{k}'}$ with eigenvalue $(-1)^{s+1}$, and so

$$\begin{aligned}\hat{S} = \frac{1}{2}\int\frac{d^3\mathbf{k}}{2k^0}\int\frac{d^3\mathbf{k}'}{2k'^0}\sum_{r,s=1,2}(-1)^{s+1}\Bigg[&\delta^{(3)}(\mathbf{k}-\mathbf{k}')\left(U_r(\mathbf{k})^\dagger U_s(\mathbf{k}'):b^\dagger_{r\mathbf{k}}b_{s\mathbf{k}'}: - V_r(\mathbf{k})^\dagger V_s(\mathbf{k}'):d_{r\mathbf{k}}d^\dagger_{s\mathbf{k}'}:\right)\\ &+\delta^{(3)}(\mathbf{k}+\mathbf{k}')\left(-e^{2ik^0t}U_r(\mathbf{k})^\dagger V_s(\mathbf{k}'):b^\dagger_{r\mathbf{k}}d^\dagger_{s\mathbf{k}'}: + e^{-2ik^0t}V_r(\mathbf{k})^\dagger U_s(\mathbf{k}'):d_{r\mathbf{k}}b_{s\mathbf{k}'}\right.\end{aligned} \tag{1.91}$$

The delta functions can be used to eliminate one of the two integrations,

$$\begin{aligned}\hat{S} = \frac{1}{2}\int\frac{d^3\mathbf{k}}{(2k^0)^2}\sum_{r,s=1,2}(-1)^{s+1}\Big[&U_r(\mathbf{k})^\dagger U_s(\mathbf{k}):b^\dagger_{r\mathbf{k}}b_{s\mathbf{k}}: - V_r(\mathbf{k})^\dagger V_s(\mathbf{k}):d_{r\mathbf{k}}d^\dagger_{s\mathbf{k}}:\\ &- e^{2ik^0t}U_r(\mathbf{k})^\dagger V_s(-\mathbf{k}):b^\dagger_{r\mathbf{k}}d^\dagger_{s-\mathbf{k}}: + e^{-2ik^0t}V_r(\mathbf{k})^\dagger U_s(-\mathbf{k}):d_{r\mathbf{k}}b_{s-\mathbf{k}}:\Big]\end{aligned} \tag{1.92}$$

Finally, using the projection relations of the polarization vectors, and then normal-order relations, we find the expected result

$$\begin{aligned}\hat{S} &= \frac{1}{2}\int\frac{d^3\mathbf{k}}{(2k^0)^2}\sum_{r,s=1,2}(-1)^{s+1}2k^0\delta_{r,s}\left(:b^\dagger_{r\mathbf{k}}b_{s\mathbf{k}}: - :d_{r\mathbf{k}}d^\dagger_{s\mathbf{k}}:\right)\\ &= \frac{1}{2}\int\frac{d^3\mathbf{k}}{2k^0}\sum_{r=1,2}(-1)^{r+1}\left(:b^\dagger_{r\mathbf{k}}b_{r\mathbf{k}}: - :d_{r\mathbf{k}}d^\dagger_{r\mathbf{k}}:\right)\\ &= \frac{1}{2}\int\frac{d^3\mathbf{k}}{2k^0}\sum_{r=1,2}(-1)^{r+1}\left(b^\dagger_{r\mathbf{k}}b_{r\mathbf{k}} + d^\dagger_{r\mathbf{k}}d_{r\mathbf{k}}\right).\end{aligned} \tag{1.93}$$

6. Let's introduce the number of particles operators $N_{r\mathbf{k}} = b^\dagger_{r\mathbf{k}} b_{r\mathbf{k}}$, $\bar{N}_{r\mathbf{k}} = d^\dagger_{r\mathbf{k}} d_{r\mathbf{k}}$, then we can show that

$$\begin{aligned}
&[N_{r\mathbf{k}}, b^\dagger_{s\mathbf{k}'}] = b^\dagger_{r\mathbf{k}} b_{r\mathbf{k}} b^\dagger_{s\mathbf{k}'} - b^\dagger_{s\mathbf{k}'} b^\dagger_{r\mathbf{k}} b_{r\mathbf{k}} = 2k^0 \delta_{r,s} \delta^{(3)}(\mathbf{k}-\mathbf{k}') b^\dagger_{r\mathbf{k}} - b^\dagger_{r\mathbf{k}} b^\dagger_{s\mathbf{k}'} b_{r\mathbf{k}} - b^\dagger_{s\mathbf{k}'} b^\dagger_{r\mathbf{k}} b_{r\mathbf{k}} = 2k^0 \delta_{r,s} \delta^{(3)}(\mathbf{k}-\mathbf{k}') b^\dagger_{r\mathbf{k}}, \\
&[N_{r\mathbf{k}}, d^\dagger_{s\mathbf{k}'}] = b^\dagger_{r\mathbf{k}} b_{r\mathbf{k}} d^\dagger_{s\mathbf{k}'} - d^\dagger_{s\mathbf{k}'} b^\dagger_{r\mathbf{k}} b_{r\mathbf{k}} = -b^\dagger_{r\mathbf{k}} d^\dagger_{s\mathbf{k}'} b_{r\mathbf{k}} - d^\dagger_{s\mathbf{k}'} b^\dagger_{r\mathbf{k}} b_{r\mathbf{k}} = 0,
\end{aligned} \tag{1.94}$$

and, in the same way, exchanging b and d,

$$[\bar{N}_{r\mathbf{k}}, b^\dagger_{s\mathbf{k}'}] = 0, \quad [\bar{N}_{r\mathbf{k}}, d^\dagger_{s\mathbf{k}'}] = 2k^0 \delta_{r,s} \delta^{(3)}(\mathbf{k}-\mathbf{k}') d^\dagger_{r\mathbf{k}}. \tag{1.95}$$

Note also that $N_{r\mathbf{k}}|0\rangle = \bar{N}_{r\mathbf{k}}|0\rangle = 0$. Then, notice that all our operators can actually be expressed in terms of these number operators,

$$P^\mu = \int \frac{d^3\mathbf{k}}{2k^0} k^\mu \sum_{r=1,2} (N_{r\mathbf{k}} + \bar{N}_{r\mathbf{k}}), \quad \hat{S} = \frac{1}{2} \int \frac{d^3\mathbf{k}}{2k^0} \sum_{r=1,2} (-)^{r+1} (N_{r\mathbf{k}} + \bar{N}_{r\mathbf{k}}), \quad Q = \int \frac{d^3\mathbf{k}}{2k^0} \sum_{r=1,2} (N_{r\mathbf{k}} - \bar{N}_{r\mathbf{k}}). \tag{1.96}$$

We deduce that the vacuum satisfies $P^\mu|0\rangle = \hat{S}|0\rangle = Q|0\rangle$, and creation operators

$$\begin{aligned}
&[P^\mu, b^\dagger_{r\mathbf{k}}] = k^\mu b^\dagger_{r\mathbf{k}}, \quad [\hat{S}, b^\dagger_{r\mathbf{k}}] = \frac{(-)^{r+1}}{2} b^\dagger_{r\mathbf{k}}, \quad [Q, b^\dagger_{r\mathbf{k}}] = b^\dagger_{r\mathbf{k}}, \\
&[P^\mu, d^\dagger_{r\mathbf{k}}] = k^\mu d^\dagger_{r\mathbf{k}}, \quad [\hat{S}, d^\dagger_{r\mathbf{k}}] = \frac{(-)^{r+1}}{2} d^\dagger_{r\mathbf{k}}, \quad [Q, d^\dagger_{r\mathbf{k}}] = -d^\dagger_{r\mathbf{k}},
\end{aligned} \tag{1.97}$$

As a result, we have

$$\begin{aligned}
P^\mu |1\bar{1} rs\rangle &= P^\mu b^\dagger_{r\mathbf{k}} d^\dagger_{s\mathbf{k}} |0\rangle = [P^\mu, b^\dagger_{r\mathbf{k}}] d^\dagger_{s\mathbf{k}} |0\rangle + b^\dagger_{r\mathbf{k}} [P^\mu, d^\dagger_{s\mathbf{k}}] |0\rangle + b^\dagger_{r\mathbf{k}} d^\dagger_{s\mathbf{k}} P^\mu |0\rangle \\
&= k^\mu b^\dagger_{r\mathbf{k}} d^\dagger_{s\mathbf{k}} |0\rangle + k^\mu b^\dagger_{r\mathbf{k}} d^\dagger_{s\mathbf{k}} |0\rangle = 2k^\mu |1\bar{1} rs\rangle.
\end{aligned} \tag{1.98}$$

In the same way, we find

$$\hat{S}|1\bar{1} rs\rangle = \frac{1}{2}\left((-1)^{r+1} + (-1)^{s+1}\right)|1\bar{1} rs\rangle, \quad Q|1\bar{1} rs\rangle = 0. \tag{1.99}$$

Thus, we find that this state has momentum $2\mathbf{k}$, energy $2k^0 = 2\sqrt{m^2 + \mathbf{k}^2}$, U(1) charge zero, and helicity that depends on r, s, more precisely

$$\hat{S}|1\bar{1}11\rangle = |1\bar{1}11\rangle, \quad \hat{S}|1\bar{1}12\rangle = 0, \quad \hat{S}|1\bar{1}21\rangle = 0, \quad \hat{S}|1\bar{1}22\rangle = -|1\bar{1}22\rangle. \tag{1.100}$$

1.4　笔记: 量子场论

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1.5　Classical Field Theory.

1.5.1　Lagrange formalism for particles.

Classical mechanics.

$L(q_n, \dot{q}_n)$ Lagrange function [assume no explicit time dependence], $q_n, n = 1, \cdots, N$ generalized coordinates.

$$S[q_n] = \int_{t_1}^{t_2} L(q_n(t), \dot{q}_n(t))dt$$

Action Functional of orbits $q_n(t)$ with fixed starting and end points $q_n(t_1, q_n(t_2)$.

Equation of motion follows from the "action principle". The orbit taken by the physical system is the one where the action is stationary:

$$\delta S[q_n] = 0.$$

This implie the Euler-Lagrange equations

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_n} - \frac{\partial L}{\partial q_n} = 0.$$

Transition to the Hamiltonian:

Canonically conjugate momenta

$$p_n \equiv \frac{\partial L}{\partial \dot{q}_n}. \tag{1.101}$$

Then

$$H(q_n, p_n) \equiv \sum_n p_n \dot{q}_n - L(q_n, \dot{q}_n)$$

where $\dot{q}_n = \dot{q}_n(q_n, p_n)$ by solving (1) for $\dot{q}_n$.

The canonical coordinates fulfill the equations

$$\{q_n p_m\}_p = \delta_{nm} \qquad \{q_n, q_m\}_p = \{p_n, p_m\} = 0.$$

$$\begin{cases} \dot{q}_n = \{q_n, H\}_p = \frac{\partial H}{\partial p_n} \\ \dot{p}_n = \{p_n, H\}_p = -\frac{\partial H}{\partial q_n} \end{cases},$$

where $\{A, B\}_p = \frac{\partial A}{\partial q_n}\frac{\partial B}{\partial p_n} - \frac{\partial A}{\partial p_n}\frac{\partial B}{\partial q_n}$.

Canonical quantization.

$L(q_n, \dot{q}_n)$ given. Interpret $q_n, \dot{q}_n, p_n$ as operators on a Hilbert space.

Impose the commutation relations [$\hbar \equiv 1$],

$$[q_n, p_m] = i\delta_{nm} \qquad [q_n, q_m] = [p_n, p_m] = 0$$

at a fixed time.

The time evolution of the operators is

$$\dot{q}_n = \frac{1}{i}[q_n, H].$$

Note: simply $\{\ ,\ \} \to \frac{1}{i}[\ ,\ ]$.

1.5.2 Lagrange-formalism for fields.

A field is a quantity defined at every point of space and time $(\vec{x}, t)$

$$\phi(\vec{x}, t).$$

Note: the concept of position has been relegated from a dynamical variable in classical mechanism to a mere label in field theory.

例 1.5.1. *E-M field.*

$$\begin{cases} \textit{Electric field}: & \vec{E}(\vec{x}, t) \\ \textit{Magnetic field}: & \vec{B}(\vec{x}, t) \end{cases} : \qquad \textit{spatial } 3\textit{-vectors},$$

derive these two 3*-vectors from a single* 4*-component filed*

$$A^{\mu}(\vec{x}, t) = (\phi, \vec{A}) \qquad [\mu = 0, 1, 2, 3],$$

where A^{μ} *is a vector in spacetime.*

The electric and magnetic fields are

$$\vec{E} = -\nabla\phi - \frac{\partial \vec{A}}{\partial t} \qquad \textit{and} \qquad \vec{B} = \nabla \times \vec{A}.$$

The dynamics of the field is governed by a Lagrangian.

In relativistic QFT one assumes that L is an integral over Lagrangian density

$$\begin{aligned} L &= \int d^3\vec{x}\mathcal{L}(\phi_n(x), \nabla\phi_n(x), \dot{\phi}_n(x)) \\ &= \int d^3\vec{x}\mathcal{L}(\partial_\mu\phi_n(x), \phi_n(x)). \end{aligned}$$

It is difficult to construct non-local Lagrangians, which are compatible with relativistic causality.

E-L equation:

$$\begin{aligned} \delta S &= \delta \int d^4x\mathcal{L}(\phi_n, \partial_\mu\phi_n) \\ &= \int d^4x[\frac{\partial\mathcal{L}}{\partial\phi_n}\delta\phi_n + \frac{\partial\mathscr{L}}{\partial(\partial_\mu\phi_n)}\delta(\partial_\mu\phi_n)] \end{aligned}$$

Integration by parts and assume $\phi(x) \to 0$ for $|\vec{x}| \to 0$, we get

$$\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_n)}\right) - \frac{\partial\mathcal{L}}{\partial\phi_n} = 0.$$

Then we define canonically conjugated fields:

$$\pi_n(x) \equiv \frac{\partial\mathcal{L}}{\partial(\partial_0\phi_n(x))}.$$

Hamilton density:

$$H = \sum_n \int d^3x\pi_n(x)\partial_0\phi_n(x) - \int d^3x\mathcal{L} \equiv \int d^3x\mathcal{H}(x)$$

$$\Rightarrow \mathcal{H}(x) = \sum_n \pi_n \partial_0 \phi_n - \mathcal{L}.$$

Suppose space were discrete with points $\vec{x}$ on a lattice, then there is a set $\phi_{n,\vec{x}(t)}$ of canonical coordi-

$$q_n(t) \longrightarrow \phi_{n,\vec{x}}(t) \longrightarrow \phi_n(t,\vec{x}$$

$$\sum_n \longrightarrow \sum_{n,\vec{x}} \longrightarrow \sum_n$$

nates, and the previous discussion applie with substitutions

$$\delta_{nm} \longrightarrow \delta_{nm}\delta_{\vec{x}\vec{y}} \longrightarrow \delta_{nm}\delta^{(3}$$

$$L(q_n(t).\dot{q}_n(t)) \longrightarrow L[\phi_n(t), \dot{\phi}_n(t)] \longrightarrow \mathcal{L}(\phi_n(x)$$

Canonical quantization.

Impose the equal-time commutation relations

$$[\phi_n(t,\vec{x}), \pi_m(t,\vec{y}] = i\delta_{nm}\delta^{(3)}(\vec{x}-\vec{y}),$$

$$[\phi_n(t,\vec{x}), \phi_m(t,\vec{y})] = [\pi_n(t,\vec{x}), \pi_m(t,\vec{y})] = 0.$$

The E-L equation is equivalent to the Heisenberg equation

$$\dot{\phi}_n(t,\vec{x}) = \frac{1}{i}[\phi_n(t,\vec{x}), H]$$

for the field operator. The formal solution is

$$\phi_n(t,\vec{x}) = e^{iHt}\phi_n(0,\vec{x})e^{-iHt},$$

since H does not explicitly depend on t.

注 **1.5.1.** *If the commutation relations are imposed at one time, they are preserved at all times.*

$$\begin{aligned}[\phi_n(t,\vec{x}), \pi_m(t,\vec{x})] &= [e^{iH(t-t_0)}\phi_n(t_0,\vec{x})e^{-iH(t-t_0)}, e^{iH(t-t_0)}\pi_m(t_0,\vec{x})e^{-iH(t-t_0)}] \\ &= e^{iH(t-t_0)}[\phi_n(t_0,\vec{x}), \pi_m(t_0,\vec{x})]e^{-iH(t-t_0)} \\ &= i\delta_{nm}\delta^{(3)}(\vec{x}-\vec{y}).\end{aligned}$$

Unless otherwise mentioned one uses the Heisenberg picture in QFT, where the time-dependence is in the operators, not the states.

例 **1.5.2.** *The Klein-Gorden Equation.*

Consider L for a real scalar field $\phi(\vec{x},t)$.

$$\mathcal{L} = \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - \frac{1}{2}m^2\phi^2 = \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}(\vec{\nabla}\phi)^2 - \frac{1}{2}m^2\phi^2,$$

where we use the Minkowski space metric $g^{\mu\nu} = g_{\mu\nu} = \mathrm{diag}(+1,-1,-1,-1)$.

We identify the kinetic energy of the field as

$$T = \int d^3x \frac{1}{2}\dot{\phi}^2,$$

the potential energy as

$$V = \int d^3x[\frac{1}{2}(\vec{\nabla}\phi)^2 + \frac{1}{2}m^2\phi^2],$$

where we call the first term by gradient energy and the second term by "true" potential energy.

To determine the equation of motion (EOM) for ϕ, we compute

$$\frac{\partial\mathcal{L}}{\partial\phi} = -m^2\phi \qquad \textit{and} \qquad \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} = \partial^\mu\phi \equiv (\dot{\phi}, -\nabla\phi).$$

The E-L euqation is then

$$\ddot{\phi} - \nabla^2\phi + m^2\phi = 0 \qquad \textit{or} \qquad \partial_\mu\partial^\mu\phi + m^2\phi,$$

Klein-Gorden Equation.

If we consider the Lagrangian with general potential $V(\phi)$,

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi) \;\Rightarrow\; \partial_\mu\partial^\mu\phi + \frac{\partial V}{\partial\phi} = 0.$$

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例 1.5.3. *Maxwell's Equation.*

From the Lagrangian $\mathcal{L} = -\frac{1}{2}(\partial_\mu A_\nu)(\partial^\mu A^\nu) + \frac{1}{2}(\partial_\mu A^\mu)^2$.

Note: $\mathcal{L}$ has no kinetic term $\dot{A}_0^2$ for A_0.

Notice that $\frac{\partial(\partial_\alpha A_\beta)}{\partial(\partial_\mu A_\nu)} = \delta^\mu_\alpha\delta^\nu_\beta$, we compute

$$\begin{aligned}
\frac{\partial\mathcal{L}}{\partial(\partial_\mu A_\nu)} &= \frac{\partial}{\partial(\partial_\mu A_\nu)}[-\frac{1}{2}\partial_\alpha A_\beta\partial_\rho A_\sigma g^{\rho\alpha}g^{\sigma\beta} + \frac{1}{2}\partial_\alpha A_\beta\partial_\rho A_\sigma g^{\beta\alpha}g^{\sigma\rho}] \\
&= -\frac{1}{2}\delta^\mu_\alpha\delta^\nu_\beta\partial_\rho A_\sigma g^{\rho\alpha}g^{\sigma\beta} - \frac{1}{2}\partial_\alpha A_\beta\delta^\mu_\rho\delta^\nu_\sigma g^{\rho\alpha}g^{\sigma\beta} + \frac{1}{2}\delta^\mu_\alpha\delta^\nu_\beta\partial_\rho A_\sigma g^{\beta\alpha}g^{\sigma\rho} + \frac{1}{2}\partial_\alpha A_\beta\delta^\mu_\rho\delta^\nu_\sigma g^{\beta\alpha}g^{\sigma\rho} \\
&= -\frac{1}{2}\partial^\mu A^\nu - \frac{1}{2}\partial^\mu A^\nu + \frac{1}{2}\partial_\rho A^\rho g^{\mu\nu} + \frac{1}{2}\partial_\alpha A^\alpha g^{\mu\nu} \\
&= -\partial^\mu A^\nu + \partial_\rho A^\rho g^{\mu\nu}.
\end{aligned}$$

Then we have

$$\begin{aligned}
\partial_\mu[\frac{\partial\mathcal{L}}{\partial(\partial_\mu A_\nu)}] &= -\partial_\mu(\partial^\mu A^\nu) + \partial^\nu(\partial_\rho A^\rho) \\
&= -\partial_\mu[\partial^\mu A^\nu - \partial^\nu A^\mu] \equiv -\partial_\mu F^{\mu\nu}
\end{aligned}$$

where the field strength is defined by

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Using $F^{\mu\nu}$, the Maxwell Lagrange is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}.$$

1.5.3 Noether's theorem for fields

The Noether theorem establishes a relation between symmetries and conserved quantities.

There is no difference between classical and quantum field theory; $\varphi(x)$ can be a classical field or quantum field operator.

$$S[\varphi] = \int d^4\mathcal{L}(\varphi, \partial_\mu\varphi).$$

A symmetry is a transformation of the fields, which leaves action invariant

$$\varphi'(x) = \varphi(x) + \varepsilon F(x),$$

where ε is a constant and a global symmetry, F depends on $\varphi(x)$ and $\partial_\mu\varphi(x)$.

$$0 = \delta S = \varepsilon \int d^4x \frac{\delta S[\varphi]}{\delta\varphi(x)} F(x)$$

before the EOM are satisfied.

例 1.5.4. *Translation.*

$$\varphi'(x) = \varphi(x+a) = \varphi(x) + a^\mu\partial_\mu\varphi(x) + \cdots,$$

then $a^\mu \sim \varepsilon$ *and* $\partial_\mu\varphi \sim F$.

定理 1.5.1. *Noether theorem.*

For every continuous symmetry there is a conserved current

$$j^\mu(x) = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi(x))} F(x) - \kappa^\mu(x) \qquad [\partial_\mu j^\mu(x) = 0]$$

and charge

$$Q \equiv \int d^3x j^0(x) \qquad [\frac{dQ}{dt} = 0].$$

Derivation. Since the action $S[\varphi(x)]$ is invariant by assumption, $\mathcal{L}$ can only change by a total derivative, i.e.

$$\delta S = \varepsilon \int d^4x \partial_\mu\kappa^\mu(x) \tag{1.102}$$

for some $\kappa^\mu(x)$ (In particular $\kappa^\mu \equiv 0$ if $\mathcal{L}$ is invariant, not only $S[\varphi]$).

One the other hand,

$$\begin{aligned}
\delta S &= \int d^4x \frac{\partial\mathcal{L}}{\partial\varphi}\delta\varphi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi)}\delta(\partial_\mu\varphi) \\
&= \int d^4x \frac{\partial\mathcal{L}}{\partial\varphi}\varepsilon F(x) + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi)}\varepsilon\partial_\mu F(x) \\
&= \int d^4x \varepsilon F(x)\partial_\mu[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi)}] + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi)}\varepsilon\partial_\mu F(x) \\
&= \int d^4x \varepsilon\partial_\mu[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\varphi)} F].
\end{aligned} \tag{1.103}$$

Taking (3) minus (1) gives

$$\partial_\mu j^\mu(x) = 0$$

for $j^\mu(x)$ as defined above. Also,

$$\frac{dQ}{dt} = \int d^3x \partial^0 j^0(x) = -\int d^3x \vec{\nabla} \cdot \vec{j}(x) = 0,$$

Assuming as usual that the fields vanish as $|\vec{x}| \to \infty$. □

注 1.5.2. *The conservation of the current is only true for the fields satisfying the filed equations, because the E-L equation was used to obtain (3). In particular, in the quantum theory, this will not be true of every field configuration we sum over in the path integral, only the saddle-point configurations.*

Translation and the Energy-Momentum Tensor.

Consider the infinitesimal translation $x^\nu \to x^\nu - \varepsilon^\nu$, then

$$\phi_n(x) \to \phi_n(x) + \varepsilon^\nu \partial_\nu \phi_n(x) \mathcal{L}(x) \to \mathcal{L}(x) + \varepsilon^\nu \partial_\nu \mathcal{L}(x)$$

rewrite $\varepsilon^\nu \partial_\nu \mathcal{L}(x)$ into $\varepsilon^\nu \partial_\mu [\delta^\mu_\nu \mathcal{L}(x)]$.

Then four conserved currents $(j^\mu)_\nu$, one for each of ε^ν, $\nu = 0, 1, 2, 3$,

$$(j^\mu)_\nu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n - \delta^\mu_\nu \mathcal{L}(x) \equiv T^\mu{}_\nu,$$

where $T^\mu{}_\nu$ denote the Energy-Momentum tensor satisfying $\partial_\mu T^\mu{}_\nu = 0$.

Four conserved quantities are given by

$$\text{the total energy of the field configuration : } E = \int d^3x T^{0,0},$$

$$\text{the total momentum : } p^i = \int d^3x T^{0,i}.$$

例 1.5.5. *The scalar field theory:* $\mathcal{L} = \frac{1}{2}\partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2$. $T^{\mu\nu} = \partial^\mu \phi \partial^\nu \phi - g^{\mu\nu}\mathcal{L}$. *Using EOM for* ϕ, $(\partial^2 + m^2)\phi = 0$, *then* $\partial_\mu T^{\mu\nu} = 0$.

The conserved energy $E = \int d^3x \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\vec{\nabla}\phi)^2 + \frac{1}{2}m^2\phi^2$ *and the total momentum* $p^i = \int d^3x \dot{\phi} \partial^i \phi$.

1.5.4 Lorentz transformation

Lorentz invariant is symmetry under rotations and boosts.

warning|I ignore some introductions to math concepts.

Among the most important symmetries of relativistic QFT are those which arise from the Lorentz transformations themselves.

The spacetime inteval $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$, where $g_{\mu\nu} = g^{\mu\nu} = \text{diag}(1, -1, -1, -1)$.

Any coordinate transformation $x^\mu \mapsto x'^\mu$ that satisfies

$$g_{\mu\nu} dx'^\mu dx'\nu = g_{\mu\nu} dx^\mu dx^\nu \qquad \text{or} \qquad g_{\mu\nu} \frac{\partial x'^\mu}{\partial x^\rho} \frac{\partial x'^\nu}{\partial x^\sigma} = g_{\rho\sigma}$$

is linear

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu + a^\nu$$

where a^μ are arbitrary constants and Λ is a constant matrix satifying

$$g_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma = g_{\rho\sigma} \qquad \text{or} \qquad \Lambda^T G\Lambda = G \tag{1.104}$$

These transformations form a group.

It's easy to check the ransformation satisfies

$$T(\overline{\Lambda},\overline{a})T(\Lambda,a) = T(\overline{\Lambda}\Lambda, \overline{\Lambda}a + \overline{a}).$$

Taking the determinant of (4) gives $(\det\Lambda)^2 = 1$, so $\Lambda^\mu{}_\nu$ has an inverse

$$(\Lambda^{-1})^\rho{}_\nu = \Lambda_\nu{}^\rho \equiv g_{\nu\mu}g^{\rho\sigma}\Lambda^\mu{}_\sigma$$

The whole group of $T(\Lambda,a)$ is **Poincaré group**.

If $a^\mu = 0$, $T(\Lambda,0)$ is known as **homogeneous Lorentz group**.

例 1.5.6. *Some examples.*

A rotation by θ about the x^3-axis,

$$\Lambda^\mu{}_\nu = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

A boost by v along the x^1-axis,

$$\Lambda^\mu{}_\nu = \begin{pmatrix} \gamma & -\gamma v & 0 & 0 \\ -\gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

with $\gamma = \frac{1}{\sqrt{1-v^2}}$.

注 1.5.3. *Poincaré group not only acts on spacetime coordinates, but also acts on the Hilbert space of a physical system, so it is not merely represented as 4×4 matrices.*

L has four connected components.

Note that $\det\Lambda$ and $\operatorname{sgn}\Lambda^0{}_0$ are both continuous functions of $\Lambda^\mu{}_\nu$. Furthermore, $\det\Lambda = \pm 1$ and $\Lambda^0{}_0 \geq 1$ or ≤ -1. Thus $\det\Lambda$ and $\operatorname{sgn}\Lambda^0{}_0$ must be constant on any one component. The four possibilities are

$$L_+^\uparrow: \qquad \det\Lambda = 1, \qquad \operatorname{sgn}\Lambda^0{}_0 = 1, \qquad \text{which contains } 1,$$

$$L_-^\uparrow: \qquad \det\Lambda = -1, \qquad \operatorname{sgn}\Lambda^0{}_0 = 1, \qquad \text{which contains } I_s,$$

$$L_+^\downarrow: \qquad \det\Lambda = 1, \qquad \operatorname{sgn}\Lambda^0{}_0 = -1, \qquad \text{which contains } I_{st},$$

$$L_-^\downarrow: \qquad \det\Lambda = -1, \qquad \operatorname{sgn}\Lambda^0{}_0 = -1, \qquad \text{which contains } I_t,$$

The Lorentz transformation:

Space inversion I_s:

$$I_s = \text{diag}(1, -1, -1, -1).$$

Time inversion I_t:

$$I_t = \text{diag}(-1, 1, 1, 1).$$

Space-time inversion I_{st}:

$$I_{st} = \text{diag}(-1, -1, -1, -1) = -I.$$

Clearly, $L_-^\uparrow \xrightarrow{I_s} L_+^\uparrow$, $L_-^\downarrow \xrightarrow{I_t} L_+^\uparrow$ and $L_+^\downarrow \xrightarrow{I_{st}} L_+^\uparrow$.

The important subgroups of L:

The orthochronous group $L^\uparrow = L_+^\uparrow \cup L_-^\uparrow$, ${\Lambda^0}_0 \geq 1$.

The proper Lorentz group $L_+ = L_+^\uparrow \cup L_+^\downarrow$, $\det\Lambda = +1$.

The orthochorous Lorentz group $L_0 = L_+^\uparrow \cup L_-^\downarrow$, $\text{sgn}\,{\lambda^0}_0 \det\Lambda = 1$.

The proper orthochronous Lorentz group or the restricted Lorentz group $L_+^\uparrow$.

1.5.5 Classical field theory

Scalar field: $\phi(x) \xrightarrow{\Lambda} \phi'(x) = \phi(\Lambda^{-1}x)$.

Vector field: $A^\mu(x) \xrightarrow{\Lambda} {\Lambda^\mu}_\nu A^\nu(\Lambda^{-1}x)$

例 1.5.7. *The Klein-Gorden field.*

$$\partial_\mu\phi(x) \to \partial_\mu\phi(\Lambda^{-1}x) = {(\Lambda^{-1})^\nu}_\mu[\partial_\nu\phi](\Lambda^{-1}x) = {\Lambda_\mu}^\nu(\partial_\nu\phi)(\Lambda^{-1}x).$$

$$\frac{\partial}{\partial x^\mu} = \frac{\partial}{\partial y^\nu}\frac{\partial y^\nu}{\partial x^\mu} = \frac{\partial}{\partial y^\nu}{(\Lambda^{-1})^\nu}_\mu,\ y^\nu = {(\Lambda^{-1})^\nu}_\mu x^\mu.$$

The derivate term in the Lagrangian transform as

$$\begin{aligned}\mathcal{L}_{deriv}(x) = \partial_\mu\phi(x)\partial_\nu\phi(x)g^{\mu\nu} &\to {\Lambda_\mu}^\rho{\Lambda_\nu}^\sigma[\partial_\rho\phi](y)[\partial_\sigma\phi](y)g^{\mu\nu} \\ &= g^{\rho\sigma}[\partial_\rho\phi](y)[\partial_\sigma\phi](y) \\ &= \mathcal{L}_{deriv}(y).\end{aligned}$$

The potential terms transform in the same way $\phi^2(x) \to \phi^2(y)$, then

$$\begin{aligned}S = \int d^4x\mathcal{L}(x) \to \int d^4x\mathcal{L}(y) &= \int d^4y\left|\frac{\partial x}{\partial y}\right|\mathcal{L}(y) \\ &= \int d^4y\mathcal{L}(y), \qquad \text{since } |\det\Lambda| = 1.\end{aligned}$$

Quantum Lorentz Transformation and the Poincaré Algebra

For an infinitesimal inhomogeneous Lorentz transformation

$${\Lambda^\mu}_\nu = {\delta^\mu}_\nu + {\omega^\mu}_\nu,\ a^\mu = \varepsilon^\mu.$$

Then we have

$$\begin{aligned} g_{\mu\nu} =& g_{\rho\sigma}(\delta^{\rho}{}_{\mu} + \omega^{\rho}{}_{\mu})(\delta^{\sigma}{}_{\nu} + \omega^{\sigma}{}_{\nu}) \\ =& g_{\mu\nu} + \omega_{\mu\nu} + \omega_{\nu\mu} + o(\omega^2). \end{aligned}$$

Hence $\omega_{\mu\nu} = -\omega_{\nu\mu}$ is antisymmetry, hence has 6 independent components and 4 components of ε^{μ}.

An Poincaré transformation is described by $6 + 4 = 10$ parameters.

The transformation $T(\Lambda, a)$ induce a unitary linear transformation in the Hilbert space

$$|\psi\rangle \to U(\Lambda, a)|\psi\rangle.$$

The operators U satisfy

$$U(\overline{\Lambda}, \overline{a})U(\Lambda, a) = U(\overline{\Lambda}\Lambda, \overline{\Lambda}a + \overline{a}.$$

Choose $U(1, 0) = 1$ the unit operator, then

$$U(1 + \omega, \varepsilon) = 1 + \frac{1}{2} i\omega_{\rho\sigma} J^{\rho\sigma} - i\varepsilon_{\rho} P^{\rho} + \cdots$$

1.6　试卷: 量子场论

原网页

2025 秋 SIMIS 量子场论 (一) 期末试卷

Exercise 1. (12 points) *In this exercise, we consider a massive real scalar field, with the mode decomposition*

$$\Phi(x) = \frac{1}{(2\pi)^{3/2}} \int \frac{d^3\mathbf{k}}{2k^0} (a_{\mathbf{k}} e^{-ikx} + a^{\dagger}_{\mathbf{k}} e^{ikx}), \quad [a_{\mathbf{k}}, a^{\dagger}_{\mathbf{k}'}] = 2k^0 \delta^{(3)}(\mathbf{k} - \mathbf{k}'), \tag{1.105}$$

and $[a_{\mathbf{k}}, a_{\mathbf{k}'}] = [a^{\dagger}_{\mathbf{k}}, a^{\dagger}_{\mathbf{k}'}] = 0$.

1. *Show that the operator* $\Phi(x)$ *satisfies the Klein-Gordon equation. (1 point)*
2. *Compute the commutator* $[\Phi(x), : \Phi(y)\Phi(y) :]$. *(1 point)*
3. *Compute the following correlator, (2 points)*

$$\langle 0| : \Phi(x_1)\Phi(x_2)\Phi(x_3) :: \Phi(x_4)\Phi(x_5)\Phi(x_6) : |0\rangle. \tag{1.106}$$

4. *We consider the following 2-particles state*

$$|2\rangle = \frac{1}{\sqrt{2}} \int \frac{d^3\mathbf{k}_1}{2k_1^0} \frac{d^3\mathbf{k}_2}{2k_2^0} f(\mathbf{k}_1) f(\mathbf{k}_2) a^{\dagger}_{\mathbf{k}_1} a^{\dagger}_{\mathbf{k}_2} |0\rangle, \quad \text{with} \quad \int \frac{d^3\mathbf{k}}{2k^0} |f(\mathbf{k})|^2 = 1. \tag{1.107}$$

Compute the energy E_2 *and energy dispertion* ΔE_2 *of this state,*

$$E_2 = \frac{\langle 2|H|2\rangle}{\langle 2|2\rangle}, \quad \Delta E_2 = \sqrt{\frac{\langle 2|H^2|2\rangle}{\langle 2|2\rangle} - (E_2)^2} \tag{1.108}$$

where H is the free field Hamiltonian $H = \frac{1}{2}\int d^3\mathbf{k}\, a^\dagger_{\mathbf{k}} a_{\mathbf{k}}$. Express the result in terms of the moments

$$\mu_n = \int \frac{d^3\mathbf{k}}{2k^0}(k^0)^n |f(\mathbf{k})|^2. \tag{1.109}$$

(3 points)

5. *We introduce the parity operator*

$$P = \exp\left(-\frac{i\pi}{2}\int \frac{d^3\mathbf{k}}{2k^0}(a^\dagger_{\mathbf{k}} a_{\mathbf{k}} - \eta_p a^\dagger_{\mathbf{k}} a_{-\mathbf{k}})\right), \tag{1.110}$$

where $\eta_p = \pm 1$ is the intrinsic parity of the field. Show that the Hamiltonian commutes with P. (2 points)

6. *Show that the field $\Phi(t,\mathbf{x})$ transforms under parity as $P\Phi(t,\mathbf{x})P^{-1} = \Phi(t,-\mathbf{x})$. (3 points)*

Exercise 2. (10 points) *In this exercise, we study spinor QED, denote the massive Dirac field $\Psi(x)$, and the massless vector field $A_\mu(x)$.*

1. *We consider the following operator appearing in Dyson's perturbative series,*

$$\mathcal{O}_3(x_1,x_2,x_3) = (ie)^3 : \overbrace{\bar\Psi^+(x_1)\gamma^{\mu_1}A^-_{\mu_1}(x_1)\,\overbrace{\Psi^+(x_1)\bar\Psi^+(x_2)}\,\gamma^{\mu_2}A^-_{\mu_2}(x_2)\Psi^-(x_2)}^{}\,\overbrace{\bar\Psi^-(x_3)\gamma^{\mu_3}A^+_{\mu_3}(x_3)\Psi^-(x_3)} : \tag{1.111}$$

Write down the corresponding Feynman diagram assuming $x_1^0 < x_2^0 < x_3^0$. Indicate the type of particle on the diagram. (2 points)

2. *Write down the operator $\mathcal{O}_3(x_1,x_2,x_3)$ in terms of Feynman propagators and a normal ordered product. (3 points)*

3. *We consider an initial state with two photons of momentum $\mathbf{p}_1, \mathbf{p}_2$ and polarization λ_1, λ_2, and a final state with a single photon of momentum $\mathbf{q}$ and polarization ρ, namely*

$$|i\rangle = a^{(\lambda_1)\dagger}_{\mathbf{p}_1} a^{(\lambda_2)\dagger}_{\mathbf{p}_2}|0\rangle, \quad |f\rangle = a^{(\rho)\dagger}_{\mathbf{q}}|0\rangle. \tag{1.112}$$

Compute the following matrix element, (3 points)

$$\langle f| : A^-_{\mu_1}(x_1)A^-_{\mu_2}(x_2)A^+_{\mu_3}(x_3) : |i\rangle. \tag{1.113}$$

4. *Deduce the corresponding S-matrix element*

$$S^{[3]}_{fi} = \frac{1}{6}\int d^4x_1 d^4x_2 d^4x_3 \langle f|\mathcal{O}_3(x_1,x_2,x_3)|i\rangle, \tag{1.114}$$

and show that it can be written in the form

$$S^{[3]}_{fi} = \delta^{(4)}(q - p_1 - p_2)\sum_{\sigma\in S_2} \mathcal{P}^{(\lambda_{\sigma(1)},\lambda_{\sigma(2)},\rho)}_{\mu_1,\mu_2,\mu_3}(\mathbf{p}_{\sigma(1)},\mathbf{p}_{\sigma(2)})\int d^4k\, R^{\mu_1,\mu_2,\mu_3}(p_{\sigma(1)},p_{\sigma(2)},k) \tag{1.115}$$

where $\mathcal{P}^{(\lambda_1,\lambda_2,\rho)}_{\mu_1,\mu_2,\mu_3}(\mathbf{p}_1,\mathbf{p}_2)$ denotes the contribution of the polarization, and $R^{\mu_1,\mu_2,\mu_3}(p_1,p_2,k)$ is a rational function of its arguments. What is the interpretation of the remaining integral over k? Explain why this process must be discarded as non-physical. (4 points)